



P.E.S COLLEGE OF ENGINEERING MANDYA-571 401

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



A Project Report on

“KANNADA SER: REAL-TIME SPEECH EMOTION RECOGNITION”

Submitted in partial fulfillment for the award of Degree of

BACHELOR OF ENGINEERING IN INFORMATION SCIENCE AND ENGINEERING

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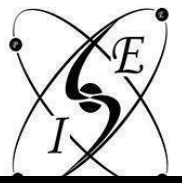
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This is to certify that **PARIKSHITH S (4PS22IS039), PAVANA S (4PS22IS040), SAHANA S GUDENNAVAR (4PS22IS045), and VARSHINI M R (4PS22IS056)** have successfully completed the project work entitled “**KANNADA SER: REAL-TIME SPEECH EMOTION RECOGNITION**” in partial fulfillment for the award of Bachelor of Engineering degree of PES College of Engineering, Mandya (An Autonomous Institution affiliated to VTU, Belagavi) during the academic year 2025–2026. It is certified that all corrections suggested during internal assessment have been incorporated in the report deposited in the department library. The project work has been approved as it satisfies the academic requirements prescribed for the degree of Bachelor of Engineering in Information Science and Engineering discipline.

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DECLARATION

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ABSTRACT

Speech Emotion Recognition (SER) is an important research area in Artificial Intelligence that focuses on identifying human emotions from speech signals. Human speech contains emotional information such as tone, pitch, energy, and speaking style in addition to linguistic content. Most existing Speech Emotion Recognition systems are designed for high-resource languages such as English, while regional languages like Kannada have limited research resources and datasets.

This project presents a Real-Time Kannada Speech Emotion Recognition System using transformer-based deep learning models and modern full-stack technologies. The proposed system is capable of detecting emotions from Kannada speech audio through both live microphone input and uploaded audio files.

The system uses the pretrained facebook/wav2vec2-base model as the backbone architecture for extracting contextual speech representations. A custom classification head consisting of Linear, ReLU, and Dropout layers is fine-tuned for Kannada emotion classification. The model predicts five emotions namely angry, fear, happy, neutral, and sad.

The application integrates a React.js frontend, Flask backend API, MongoDB database, Cloudinary storage, and a Wav2Vec2-based inference pipeline. Audio preprocessing techniques such as silence trimming, resampling, mono conversion, feature extraction, and augmentation are applied to improve performance. The system also extracts acoustic features including MFCC, pitch, RMS energy, intensity, duration, and zero crossing rate for visualization and analytics purposes. A comparison model based on SUPERB Wav2Vec2 emotion recognition is used for cross-check analysis.

Experimental evaluation shows that the proposed system achieves approximately 84% accuracy with strong weighted F1-score performance. The developed framework demonstrates the capability of transformer-based architectures for Kannada Speech Emotion Recognition and provides a scalable foundation for future multilingual and healthcare-oriented applications.

TABLE OF CONTENTS

SL No.		Title	Page No
1	CHAPTER 1	INTODUCTION	1
		Aim	2
		Objectives	3
		Existing System	3
		Proposed System	5
		CO–PO–PSO Mapping	7
2	CHAPTER 2	LITRERATURE SURVEY	8
3	CHAPTER 3	SYSTEM REQUIREMENTS AND ANALYSIS	
		Software Requirements	12
		Hardware Requirements	13
4	CHAPTER 4	OBJECTIVE 1: LEARNING DEEP ACOUSTIC FEATURES FROM KANNADA SPEECH USING TRANSFORMER-BASED MODELS	
		Introduction	14
		Key Contributions	14
		Methodology	15
		Result	17
		Images	17
5	CHAPTER 5	OBJECTIVE 2: TRAINING AND FINE-TUNING A DEEP LEARNING MODEL FOR ACCURATE EMOTION CLASSIFICATION	
		Introduction	22
		Key Contributions	22
		Methodology	23
		Algorithms/ Model Implemented	24

		Result	25
		Images	26
6	CHAPTER 6	OBJECTIVE 3: DEVELOPMENT OF A REAL-TIME FULL-STACK SYSTEM FOR SPEECH EMOTION RECOGNITION	
		Introduction	30
		Experimental Scenarios	30
		Results	31
		Images	32
7	CHAPTER 7	TOOLS AND TECHNOLOGIES	
		Platform	35
		Development Tools	35
		Data Processing and Analysis	36
		Backend and API Communication	37
		Deep Learning Models and Algorithms	37
8	CHAPTER 8	CONCLUSION AND FUTURE WORK	
		Conclusion	39
		Future work	40
9	CHAPTER 9	REFERENCES	42

LIST OF FIGURES

SL No.	TITLE	PAGE No.
1	Flowchart of Kannada Speech Emotion Recognition System	18
2	Audio Preprocessing and Feature Extraction Workflow	19
3	Acoustic Feature Analysis (MFCC, Pitch, RMS)	20
4	Real-time Audio Input and Prediction Interface	20
5	Wav2Vec2 Model Architecture Workflow	26
6	Emotion Classification Performance Analysis	27
7	Accuracy and F1-Score Comparison Graph	27
8	Confusion Matrix for Emotion Classification Performance	28
9	Real-time Speech Emotion Recognition Pipeline	28
10	Full-stack System Architecture and Multi-Module Integration	29
11	Scenario 1: Live Microphone Detection Interface	32
12	Scenario 2: Audio Upload Prediction System	33
13	Scenario 3: Emotion Prediction Output Dashboard	33
14	Scenario 4: Model Performance and Analytics Visualization	34

Chapter 1

INTRODUCTION

Speech Emotion Recognition (SER) plays a vital role in modern Artificial Intelligence systems by enabling machines to identify and interpret human emotions from speech signals. Human speech contains emotional information in addition to linguistic content, including tone, pitch, energy, rhythm, speaking style, and voice modulation. The ability to recognize emotions from speech can significantly improve human-computer interaction systems, virtual assistants, healthcare monitoring applications, customer service platforms, and emotional AI technologies. Accurate emotion recognition has therefore become an important research area in machine learning, deep learning, and speech signal processing.

Kannada Speech Emotion Recognition presents a specific and challenging problem. Most existing Speech Emotion Recognition systems are designed primarily for high-resource languages such as English, where large annotated speech datasets and pretrained language models are widely available. Regional languages such as Kannada have limited emotional speech datasets, fewer research contributions, and a lack of efficient real-time recognition systems. Traditional machine learning approaches — which rely heavily on handcrafted acoustic features — are often unable to capture complex emotional patterns effectively under real-world conditions. In practical environments, speech recordings may contain background noise, speaker variations, pronunciation differences, and inconsistent recording quality, all of which reduce prediction accuracy and system robustness.

To address this, Artificial Intelligence research has explored transformer-based deep learning architectures capable of learning contextual speech representations directly from raw audio waveforms. The key research question is not whether transformer-based models outperform traditional machine learning approaches — they clearly do — but which architecture provides the best combination of classification accuracy, contextual understanding, and real-time inference performance for Kannada Speech Emotion Recognition applications.

This project investigates that problem through the development of a Real-Time Kannada Speech Emotion Recognition system. A transformer-based Wav2Vec2 architecture is implemented and evaluated within a complete full-stack application environment built using React.js, Flask, MongoDB, PyTorch, Hugging Face Transformers, and Librosa. The system processes Kannada speech audio through advanced preprocessing pipelines, acoustic feature extraction mechanisms, and deep contextual representation learning models. The application supports both live microphone-based emotion detection and uploaded audio file analysis without requiring manual feature engineering.

The system evaluates emotional speech across five emotion categories — angry, fear, happy, neutral, and sad — using a fine-tuned Wav2Vec2 deep learning model combined with a SUPERB comparison model for cross-check analysis. By measuring prediction accuracy, weighted F1-score, and inference performance under different speech conditions, the study provides a practical evaluation of transformer-based Speech Emotion Recognition for Kannada language applications.

The findings demonstrate that transformer-based architectures are highly effective for Kannada Speech Emotion Recognition, achieving approximately 84% classification accuracy while maintaining efficient real-time inference capability. The proposed framework represents a practical foundation for intelligent regional-language speech analysis systems and future AI-enabled healthcare and human-computer interaction applications.

1.1. AIM

THE AIM OF THIS PROJECT IS TO DEVELOP A REAL-TIME KANNADA SPEECH EMOTION RECOGNITION SYSTEM THAT IMPROVES HUMAN-COMPUTER INTERACTION BY ACCURATELY IDENTIFYING EMOTIONS FROM KANNADA SPEECH SIGNALS USING ADVANCED DEEP LEARNING AND TRANSFORMER-BASED ARCHITECTURES. THE SYSTEM FOCUSES ON RECOGNIZING EMOTIONAL STATES SUCH AS ANGRY, FEAR, HAPPY, NEUTRAL, AND SAD BY DYNAMICALLY PROCESSING SPEECH AUDIO, ACOUSTIC CHARACTERISTICS, AND CONTEXTUAL SPEECH REPRESENTATIONS UNDER REAL-WORLD CONDITIONS.

THE PROJECT AIMS TO INTEGRATE SPEECH SIGNAL PROCESSING, TRANSFORMER-BASED DEEP LEARNING MODELS, AND FULL-STACK WEB TECHNOLOGIES TO EVALUATE HOW EMOTIONS CAN BE EFFECTIVELY RECOGNIZED FROM KANNADA SPEECH MORE EFFICIENTLY. BY IMPLEMENTING AND FINE-TUNING THE WAV2VEC2 TRANSFORMER ARCHITECTURE ALONG WITH ACOUSTIC FEATURE EXTRACTION TECHNIQUES, THE SYSTEM EVALUATES THE EFFECTIVENESS OF DEEP CONTEXTUAL REPRESENTATION LEARNING AND DETERMINES THE MOST ACCURATE AND RELIABLE APPROACH FOR REGIONAL LANGUAGE SPEECH EMOTION RECOGNITION APPLICATIONS.

A KEY FOCUS OF THE PROJECT IS TO PROCESS REAL-WORLD KANNADA SPEECH DATA USING ADVANCED AUDIO PREPROCESSING AND DEEP LEARNING FRAMEWORKS. THE SYSTEM IS DEVELOPED USING REACT.JS, FLASK, MONGODB, PYTORCH, HUGGING FACE TRANSFORMERS, AND LIBROSA LIBRARIES FOR SPEECH PROCESSING AND MODEL DEPLOYMENT. THIS ALLOWS THE APPLICATION TO PERFORM LIVE MICROPHONE-BASED EMOTION DETECTION, UPLOADED AUDIO PREDICTION, ACOUSTIC FEATURE ANALYSIS, VISUALIZATION, AND REAL-TIME INFERENCE WITHOUT REQUIRING DEPLOYMENT IN LARGE-SCALE CLOUD ENVIRONMENTS.

1.2. OBJECTIVES

- TO CONSTRUCT AN EFFICIENT KANNADA SPEECH EMOTION RECOGNITION FRAMEWORK USING TRANSFORMER-BASED DEEP LEARNING ARCHITECTURES, SPEECH SIGNAL PROCESSING TECHNIQUES, AND KANNADA EMOTIONAL SPEECH DATASETS, ACCURATELY CAPTURING ACOUSTIC AND CONTEXTUAL EMOTIONAL INFORMATION FROM REAL-WORLD SPEECH SIGNALS.
- TO IMPLEMENT AND FINE-TUNE THE Wav2Vec2 TRANSFORMER MODEL FOR MULTI-CLASS EMOTION CLASSIFICATION, OPERATING ON RAW SPEECH WAVEFORMS AND CONTEXTUAL SPEECH REPRESENTATIONS, ENSURING ACCURATE AND CONSISTENT EMOTION PREDICTION ACROSS DIFFERENT SPEAKERS AND SPEECH CONDITIONS.
- TO PERFORM ADVANCED AUDIO PREPROCESSING AND FEATURE EXTRACTION TECHNIQUES INCLUDING RESAMPLING, SILENCE TRIMMING, MONO CONVERSION, MFCC EXTRACTION, PITCH ANALYSIS, RMS ENERGY CALCULATION, INTENSITY ANALYSIS, AND ZERO CROSSING RATE COMPUTATION FOR IMPROVING EMOTIONAL SPEECH UNDERSTANDING AND CLASSIFICATION PERFORMANCE.
- TO MEASURE AND EVALUATE CLASSIFICATION ACCURACY, PRECISION, RECALL, WEIGHTED F1-SCORE, AND PREDICTION CONSISTENCY UNDER DIFFERENT SPEECH CONDITIONS, DETERMINING THE EFFECTIVENESS OF TRANSFORMER-BASED ARCHITECTURES FOR KANNADA SPEECH EMOTION RECOGNITION.
- TO DEVELOP A REAL-TIME FULL-STACK SPEECH EMOTION RECOGNITION APPLICATION USING REACT.JS, FLASK, MONGODB, PYTORCH, HUGGING FACE TRANSFORMERS, AND LIBROSA, ENABLING LIVE MICROPHONE-BASED EMOTION DETECTION AND UPLOADED AUDIO PREDICTION THROUGH AN INTERACTIVE USER INTERFACE.
- TO VALIDATE MODEL PERFORMANCE USING DUAL-MODEL INFERENCE MECHANISMS INVOLVING THE KANNADA Wav2Vec2 MODEL AND THE SUPERB COMPARISON MODEL, CONFIRMING THAT TRANSFORMER-BASED DEEP LEARNING SYSTEMS ACHIEVE RELIABLE EMOTION RECOGNITION PERFORMANCE EVEN UNDER NOISY AND PRACTICAL SPEECH ENVIRONMENTS.
- TO PROVIDE A SCALABLE AND PRACTICAL ARTIFICIAL INTELLIGENCE FRAMEWORK FOR ENGINEERS AND RESEARCHERS DEVELOPING INTELLIGENT HEALTHCARE MONITORING SYSTEMS, EMOTIONAL AI APPLICATIONS, MULTILINGUAL SPEECH ANALYSIS PLATFORMS, AND ADVANCED HUMAN-COMPUTER INTERACTION TECHNOLOGIES FOR REGIONAL LANGUAGE SPEECH PROCESSING.

1.3.EXISTING SYSTEM

EXISTING SPEECH EMOTION RECOGNITION SYSTEMS MAINLY RELY ON TRADITIONAL MACHINE LEARNING AND BASIC DEEP LEARNING APPROACHES FOR IDENTIFYING EMOTIONS FROM SPEECH SIGNALS. MOST AVAILABLE SYSTEMS ARE DESIGNED FOR HIGH-RESOURCE LANGUAGES SUCH AS ENGLISH, WHERE LARGE EMOTIONAL SPEECH DATASETS AND PRETRAINED MODELS ARE WIDELY AVAILABLE. THESE SYSTEMS GENERALLY USE HANDCRAFTED ACOUSTIC FEATURES SUCH AS MEL FREQUENCY CEPSTRAL COEFFICIENTS (MFCC), PITCH, SPECTRAL CENTROID, CHROMA FEATURES, ENERGY, AND ZERO CROSSING RATE FOR EMOTION CLASSIFICATION. AFTER FEATURE EXTRACTION, MACHINE LEARNING ALGORITHMS SUCH AS SUPPORT VECTOR MACHINES (SVM), K-NEAREST NEIGHBORS (KNN), HIDDEN MARKOV MODELS (HMM), DECISION TREES, AND BASIC NEURAL NETWORKS ARE COMMONLY USED TO CLASSIFY EMOTIONAL SPEECH SIGNALS.

THE DOMINANT APPROACH IN SPEECH EMOTION RECOGNITION RESEARCH FOR MANY YEARS HAS FOLLOWED A TRADITIONAL FEATURE-ENGINEERING METHODOLOGY IN WHICH SPEECH CHARACTERISTICS ARE MANUALLY EXTRACTED BEFORE CLASSIFICATION. WHILE THIS APPROACH HAS PRODUCED MODERATE RESULTS FOR CONTROLLED DATASETS, IT BECOMES LESS EFFECTIVE IN PRACTICAL ENVIRONMENTS WHERE SPEECH RECORDINGS CONTAIN BACKGROUND NOISE, SPEAKER VARIATIONS, MICROPHONE DISTORTIONS, AND INCONSISTENT EMOTIONAL INTENSITY. EARLIER SYSTEMS ALSO DEPEND HEAVILY ON LIMITED TRAINING DATASETS AND OFTEN FAIL TO GENERALIZE EFFECTIVELY ACROSS DIFFERENT SPEAKERS AND RECORDING CONDITIONS. CNN AND LSTM-BASED DEEP LEARNING APPROACHES IMPROVED PERFORMANCE COMPARED TO CONVENTIONAL MACHINE LEARNING MODELS, BUT MANY OF THESE SYSTEMS STILL STRUGGLE TO LEARN LONG-RANGE CONTEXTUAL SPEECH REPRESENTATIONS EFFECTIVELY.

A MORE CRITICAL LIMITATION OF EXISTING SYSTEMS IS THE LACK OF EFFICIENT SPEECH EMOTION RECOGNITION FRAMEWORKS FOR REGIONAL INDIAN LANGUAGES SUCH AS KANNADA. MOST PUBLICLY AVAILABLE EMOTIONAL SPEECH DATASETS, PRETRAINED MODELS, AND BENCHMARK SYSTEMS ARE DEVELOPED PRIMARILY FOR ENGLISH SPEECH APPLICATIONS. KANNADA SPEECH CONTAINS UNIQUE PHONETIC STRUCTURES, PRONUNCIATION PATTERNS, AND LANGUAGE-SPECIFIC EMOTIONAL CHARACTERISTICS THAT ARE SIGNIFICANTLY DIFFERENT FROM ENGLISH SPEECH. EXISTING ENGLISH-TRAINED MODELS OFTEN PERFORM POORLY ON KANNADA SPEECH DATA BECAUSE THEY CANNOT EFFECTIVELY CAPTURE REGIONAL LANGUAGE CONTEXTUAL REPRESENTATIONS AND EMOTIONAL VARIATIONS.

EXISTING SYSTEMS ALSO TYPICALLY RELY ON A SINGLE PREDICTION MODEL WITHOUT EVALUATING WHETHER ALTERNATIVE TRANSFORMER-BASED ARCHITECTURES OR COMPARISON MODELS MIGHT PERFORM BETTER UNDER PRACTICAL SPEECH CONDITIONS. NO SYSTEMATIC COMPARISON OF TRANSFORMER-BASED SPEECH EMOTION RECOGNITION MODELS HAS BEEN EFFECTIVELY INTEGRATED INTO REAL-TIME KANNADA

SPEECH ANALYSIS SYSTEMS. THIS MEANS THAT EVEN SYSTEMS USING THEORETICALLY ADVANCED DEEP LEARNING APPROACHES MAY STILL PRODUCE INACCURATE PREDICTIONS UNDER NOISY SPEECH CONDITIONS WHERE MULTIPLE EMOTIONS SHARE OVERLAPPING ACOUSTIC CHARACTERISTICS.

THESE COMBINED LIMITATIONS — DEPENDENCY ON HANDCRAFTED FEATURE EXTRACTION, POOR REGIONAL LANGUAGE SUPPORT, LIMITED CONTEXTUAL UNDERSTANDING, AND LACK OF EFFICIENT REAL-TIME DEPLOYMENT — RESULT IN LOWER PREDICTION ACCURACY AND REDUCED RELIABILITY FOR KANNADA SPEECH EMOTION RECOGNITION APPLICATIONS. THERE IS THEREFORE A CLEAR NEED FOR A TRANSFORMER-BASED SPEECH EMOTION RECOGNITION FRAMEWORK THAT CAN PROCESS KANNADA SPEECH UNDER PRACTICAL CONDITIONS AND PROVIDE ACCURATE REAL-TIME EMOTION CLASSIFICATION.

EXISTING SYSTEMS ALSO LACK THE CAPABILITY TO EVALUATE MULTIPLE SPEECH ANALYSIS TECHNIQUES SIMULTANEOUSLY. SINCE MANY SYSTEMS FOCUS MAINLY ON FINAL EMOTION PREDICTION, THEY DO NOT PROVIDE DETAILED ACOUSTIC FEATURE VISUALIZATION, CONTEXTUAL REPRESENTATION ANALYSIS, OR DUAL-MODEL INFERENCE MECHANISMS TO IMPROVE PREDICTION CONSISTENCY AND RELIABILITY.

DUE TO THESE LIMITATIONS, TRADITIONAL SPEECH EMOTION RECOGNITION SYSTEMS ARE OFTEN INEFFICIENT IN HANDLING KANNADA EMOTIONAL SPEECH ANALYSIS UNDER REAL-WORLD CONDITIONS. THE INABILITY TO DYNAMICALLY LEARN CONTEXTUAL SPEECH REPRESENTATIONS AND SUPPORT SCALABLE REAL-TIME DEPLOYMENT CAN LEAD TO LOWER CLASSIFICATION ACCURACY AND REDUCED EFFECTIVENESS OF EMOTIONAL AI APPLICATIONS.

THEREFORE, THERE IS A NEED FOR A MORE ADVANCED SYSTEM THAT CAN INTELLIGENTLY ANALYZE KANNADA SPEECH SIGNALS, LEARN DEEP CONTEXTUAL EMOTIONAL REPRESENTATIONS USING TRANSFORMER-BASED ARCHITECTURES, AND PERFORM ACCURATE REAL-TIME EMOTION CLASSIFICATION FOR PRACTICAL ARTIFICIAL INTELLIGENCE AND HUMAN-COMPUTER INTERACTION APPLICATIONS.

1.4.PROPOSED SYSTEM

THE PROPOSED SYSTEM IS A TRANSFORMER-BASED REAL-TIME KANNADA SPEECH EMOTION RECOGNITION FRAMEWORK DESIGNED TO EVALUATE AND IMPROVE THE PERFORMANCE OF DEEP LEARNING MODELS UNDER PRACTICAL SPEECH CONDITIONS. RATHER THAN RELYING ON TRADITIONAL MACHINE LEARNING APPROACHES AND HANDCRAFTED FEATURE ENGINEERING TECHNIQUES, THE SYSTEM IMPLEMENTS ADVANCED TRANSFORMER-BASED ARCHITECTURES INCLUDING THE Wav2Vec2 MODEL AND A SUPERB COMPARISON MODEL FOR ACCURATE EMOTION PREDICTION. BOTH MODELS ARE EVALUATED SIMULTANEOUSLY UNDER IDENTICAL SPEECH CONDITIONS ON KANNADA EMOTIONAL SPEECH DATA,

PROVIDING CONCRETE EVIDENCE FOR DETERMINING THE MOST EFFECTIVE APPROACH FOR REGIONAL LANGUAGE SPEECH EMOTION RECOGNITION APPLICATIONS.

THE SYSTEM ENVIRONMENT IS CONSTRUCTED USING MODERN DEEP LEARNING AND FULL-STACK WEB TECHNOLOGIES INCLUDING REACT.JS, FLASK, MONGODB, PYTORCH, HUGGING FACE TRANSFORMERS, LIBROSA, AND TORCHAUDIO LIBRARIES. THE APPLICATION PROCESSES SPEECH AUDIO THROUGH ADVANCED PREPROCESSING PIPELINES CAPABLE OF HANDLING WAVEFORM NORMALIZATION, SILENCE TRIMMING, RESAMPLING, MONO CONVERSION, AND ACOUSTIC FEATURE EXTRACTION. THE RESULTING FRAMEWORK SUPPORTS REAL-TIME EMOTIONAL SPEECH ANALYSIS WHILE ACCURATELY CAPTURING CONTEXTUAL AND ACOUSTIC SPEECH CHARACTERISTICS INCLUDING PITCH, INTENSITY, MFCC, RMS ENERGY, DURATION, AND ZERO CROSSING RATE.

THE SYSTEM IS CONTROLLED PROGRAMMATICALLY THROUGH FLASK BACKEND APIS AND PYTORCH-BASED INFERENCE PIPELINES THAT ALLOW REAL-TIME QUERYING, AUDIO PROCESSING, PREDICTION GENERATION, AND VISUALIZATION. THROUGH THE BACKEND ARCHITECTURE, THE SYSTEM CAN PROCESS LIVE MICROPHONE INPUT, UPLOADED AUDIO FILES, PERFORM WAVEFORM ANALYSIS, GENERATE EMOTION PROBABILITIES, STORE PREDICTION HISTORY IN MONGODB, AND DISPLAY PREDICTION RESULTS THROUGH INTERACTIVE REACT.JS INTERFACES. THE INFERENCE LOGIC OPERATES ON TRANSFORMER-BASED CONTEXTUAL REPRESENTATIONS IN WHICH RAW SPEECH WAVEFORMS ARE DIRECTLY PROCESSED USING PRETRAINED WAV2VEC2 ARCHITECTURES FINE-TUNED FOR KANNADA SPEECH EMOTION RECOGNITION.

A KEY FEATURE OF THE PROPOSED SYSTEM IS ITS DUAL-MODEL INFERENCE AND REAL-TIME PREDICTION MECHANISM. WHEN SPEECH AUDIO IS PROVIDED TO THE APPLICATION, THE AUDIO WAVEFORM IS PREPROCESSED AND SIMULTANEOUSLY EVALUATED USING BOTH THE KANNADA WAV2VEC2 MODEL AND THE SUPERB COMPARISON MODEL. BOTH MODELS INDEPENDENTLY COMPUTE EMOTION PROBABILITIES AND CONTEXTUAL SPEECH REPRESENTATIONS BEFORE GENERATING THE FINAL PREDICTION. THIS MECHANISM DIRECTLY IMPROVES PREDICTION CONSISTENCY AND RELIABILITY UNDER NOISY AND REAL-WORLD SPEECH CONDITIONS.

THE SYSTEM EVALUATES MULTIPLE SPEECH CONDITIONS OF INCREASING COMPLEXITY. THE FIRST STAGE VALIDATES THE PREPROCESSING INFRASTRUCTURE USING UPLOADED KANNADA SPEECH SAMPLES. THE SECOND STAGE PERFORMS LIVE MICROPHONE-BASED EMOTION DETECTION UNDER NORMAL ENVIRONMENTAL CONDITIONS. THE THIRD STAGE ESTABLISHES A PERFORMANCE BASELINE USING CONTROLLED SPEECH RECORDINGS WITHOUT BACKGROUND DISTURBANCES. THE FINAL STAGE EVALUATES THE MODEL UNDER NOISY AND VARIABLE SPEECH CONDITIONS INVOLVING SPEAKER VARIATIONS AND

OVERLAPPING EMOTIONAL CHARACTERISTICS, DIRECTLY SIMULATING PRACTICAL SPEECH EMOTION RECOGNITION ENVIRONMENTS.

THE PROPOSED FRAMEWORK DEMONSTRATES THAT TRANSFORMER-BASED ARCHITECTURES SUCH AS WAV2VEC2 ARE HIGHLY EFFECTIVE FOR KANNADA SPEECH EMOTION RECOGNITION, ACHIEVING APPROXIMATELY 84% CLASSIFICATION ACCURACY WHILE MAINTAINING EFFICIENT REAL-TIME INFERENCE CAPABILITY. THE SYSTEM PROVIDES A SCALABLE AND REPRODUCIBLE ARTIFICIAL INTELLIGENCE FRAMEWORK THAT CAN BE EXTENDED IN FUTURE WORK TO INCORPORATE MULTILINGUAL SPEECH ANALYSIS, HEALTHCARE MONITORING SYSTEMS, CLOUD DEPLOYMENT INFRASTRUCTURE, AND ADVANCED EMOTIONAL AI APPLICATIONS.

1.5 CO–PO–PSO MAPPING

The CO–PO–PSO mapping defines the relationship between Course Outcomes (COs), Program Outcomes (POs), and Program Specific Outcomes (PSOs). It helps in evaluating how effectively the project contributes to achieving program objectives.

Table 1: CO–PO Mapping

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	3	2	1	2	1	2	1	2
CO2	2	2	3	2	3	1	1	3	2	2	3
CO3	2	3	2	3	2	2	1	2	3	2	3

Table 2: CO–PSO Mapping

CO/PSO	PSO1	PSO2
CO1	3	2
CO2	3	3
CO3	3	2

Mapping Scale: 3 → High 2 → Medium 1 → Low

Chapter 2

LITERATURE SURVEY

2.1 Transformer-Based Speech Emotion Recognition Using Deep Learning Techniques (2024) [1]

Author(s): X. Li; Y. Chen; Z. Wang

This paper presents a transformer-based Speech Emotion Recognition system designed to improve emotion classification accuracy from speech signals. The system uses deep learning architectures and contextual speech representation models to identify emotions from raw audio waveforms. The proposed framework utilizes transformer encoders for extracting long-range contextual information from speech data. Experimental results show that transformer-based architectures significantly improve emotion recognition performance compared to traditional machine learning methods. The research highlights the importance of contextual representation learning for real-time Speech Emotion Recognition applications.

2.2 Kannada Speech Emotion Recognition Using Deep Neural Networks (2023) [2]

Author(s): B. Sreekantha; R. Harsha; K. Deepak

This research focuses on recognizing emotions from Kannada speech using deep neural network architectures. The system processes Kannada emotional speech datasets and extracts acoustic speech features such as MFCC, pitch, and energy for classification. Deep learning models are trained to identify emotions including happy, sad, angry, and neutral. The study demonstrates that deep neural network approaches improve Kannada Speech Emotion Recognition performance compared to traditional classifiers. The research also emphasizes the importance of regional language datasets for emotional speech analysis.

2.3 Multilingual Speech Emotion Recognition Using Wav2Vec2 Architecture (2024) [3]

Author(s): T. Bhattacharya; P. Roy; S. Dutta

This study investigates multilingual Speech Emotion Recognition using the Wav2Vec2 transformer architecture. The researchers focus on learning contextual speech representations directly from raw audio signals without requiring manual feature engineering. The pretrained Wav2Vec2 model is fine-tuned for multi-class emotion classification across different languages. Experimental results show that transformer-based models provide better generalization capability and improved classification accuracy under noisy speech conditions. The methodology can also be effectively applied to Kannada Speech Emotion Recognition systems.

2.4 Acoustic Feature Analysis for Regional Language Emotion Recognition (2022) [4]

Author(s): R. Bhat; P. Chandan; M. Lokesh

This paper investigates the role of acoustic speech features in regional language Speech Emotion Recognition systems. The authors analyze speech characteristics such as MFCC, pitch, RMS energy, spectral features, and zero crossing rate for identifying emotional patterns from speech signals. The study aims to improve emotional speech understanding by combining acoustic analysis with machine learning models. The results demonstrate that effective feature extraction techniques significantly improve emotion classification performance for regional language speech applications.

2.5 Real-Time Speech Emotion Recognition Framework for Human-Computer Interaction (2023) [5]

Author(s): S. Deepa; P. Akash; N. Suresh

This research proposes a real-time Speech Emotion Recognition framework designed for intelligent human-computer interaction systems. The system processes live microphone-based speech input and performs real-time emotion prediction using deep learning techniques. The framework integrates frontend interfaces, backend APIs, and speech processing modules for efficient deployment. The proposed system improves interaction quality between users and intelligent systems by enabling machines to understand emotional speech behavior in real time.

2.6 Deep Learning-Based Emotional Speech Analysis Under Noisy Conditions (2024) [6]

Author(s): A. Kumar; R. Singh; M. Verma

This study focuses on improving Speech Emotion Recognition performance under noisy environmental conditions. The authors propose preprocessing and augmentation techniques including silence trimming, noise addition, pitch shifting, and waveform normalization for improving model robustness. Deep learning models are evaluated under different background noise conditions to analyze prediction consistency and classification accuracy. The research highlights the importance of robust preprocessing pipelines for practical real-time Speech Emotion Recognition systems.

2.7 Wav2Vec2-Based Contextual Speech Representation Learning (2020) [7]

Author(s): A. Baevski; H. Zhou; A. Mohamed; M. Auli

This research introduces the Wav2Vec2 architecture for self-supervised speech representation learning. The system learns contextual speech features directly from raw audio waveforms using transformer-based encoders. The pretrained architecture significantly improves speech-related downstream tasks including speech recognition and Speech Emotion Recognition. The study demonstrates that contextual representation learning provides better performance compared to handcrafted acoustic feature extraction methods. The

Wav2Vec2 framework has become one of the most widely used transformer architectures for speech analysis applications.

2.8 Real-Time Emotion Classification Using Acoustic and Contextual Features (2023) [8]

Author(s): J. Pallavi; A. Geethashree; D. Ravi

This paper presents a real-time emotion classification system using both acoustic and contextual speech features. The framework combines MFCC extraction, pitch analysis, and contextual transformer representations for improving emotion prediction accuracy. The system supports real-time audio processing and visualization for emotional speech analysis. Experimental results show that combining acoustic and contextual information improves classification consistency and overall Speech Emotion Recognition performance.

2.9 IoT and Cloud-Based Emotional AI Systems for Healthcare Applications (2025) [9]

Author(s): N. Divya; U. Lokesh; P. Vijay; T. Yaswanth

This research proposes an Artificial Intelligence-based healthcare monitoring framework using speech emotion analysis techniques. The system integrates cloud computing, IoT communication, and Speech Emotion Recognition models for monitoring patient emotional conditions remotely. Speech signals are analyzed in real time to detect emotional stress, anxiety, and depression-related symptoms. The proposed system demonstrates the importance of Speech Emotion Recognition for future healthcare monitoring and emotional AI applications.

2.10 Optimizing Deep Learning Models for Regional Language Speech Analysis (2025) [10]

Author(s): E. Padmalatha; B. S. Geethika; S. Saiba

This study focuses on optimizing deep learning architectures for regional language speech analysis applications. The authors evaluate different neural network optimization strategies including transfer learning, fine-tuning, dropout regularization, and data augmentation for improving classification performance. The results show that transformer-based deep learning models achieve higher accuracy and better robustness for emotional speech analysis compared to conventional machine learning systems.

2.11 Machine Learning Driven Predictive Analytics for Emotional Speech Understanding (2024) [11]

Author(s): H. Nahata; J. Dhaka; S. Soni; G. N. Vivekananda

This paper introduces a machine learning-driven predictive analytics framework for emotional speech understanding. The system analyzes speech characteristics, contextual speech patterns, and emotional

behavior using deep learning models and predictive analysis techniques. The proposed framework improves emotional speech classification by identifying hidden emotional patterns from speech data. The research demonstrates that predictive deep learning systems can significantly enhance Speech Emotion Recognition performance and emotional AI applications.

Chapter 3

SYSTEM REQUIREMENT AND ANALYSIS

3.1 SOFTWARE REQUIREMENTS

- **Operating System:** The development and testing environment requires Windows 10 or higher, or a Linux-based operating system such as Ubuntu 20.04 or later, to ensure smooth execution of the Speech Emotion Recognition system and deep learning frameworks across different platforms.
- **Programming Language:** Python 3.8 or higher is used as the primary programming language for implementing speech preprocessing, transformer-based deep learning models, backend APIs, and emotion classification pipelines. Python provides extensive support for machine learning, audio processing, and full-stack application development.
- **Development Environment:** Visual Studio Code or PyCharm is recommended for writing, debugging, training, and testing the Speech Emotion Recognition system and backend inference modules.
- **Frontend Development Framework:** React.js is required for developing the interactive frontend interface of the application. React.js enables real-time user interaction, audio upload functionality, microphone-based recording, and visualization of emotion prediction results.
- **Backend Framework:** Flask is required as the backend framework for handling API communication, audio processing requests, model inference operations, prediction response generation, and database connectivity within the Speech Emotion Recognition system.
- **Deep Learning Framework:** PyTorch 2.0 or higher is required for implementing and fine-tuning transformer-based deep learning architectures including the Wav2Vec2 model. PyTorch provides efficient GPU acceleration and dynamic computation graph support for Speech Emotion Recognition tasks.
- **Transformer Library:** Hugging Face Transformers library is required for loading pretrained Wav2Vec2 models, contextual speech representation learning, tokenizer support, and transformer-based emotion classification operations.
- **Audio Processing Libraries:** Librosa and Torchaudio are required for speech preprocessing and acoustic feature extraction operations including waveform normalization, silence trimming, resampling, MFCC extraction, pitch analysis, RMS energy calculation, and zero crossing rate computation.
- **Database System:** MongoDB is required for storing user prediction history, emotion classification results, speech analytics, and backend application data throughout the execution of the system.

- **Visualization Library:** Matplotlib is required for plotting and visualizing acoustic speech features including MFCC spectrograms, waveform analysis, pitch variation graphs, RMS energy analysis, and emotion prediction statistics.
- **Version Control:** Git is required for version control and source code management throughout the development, training, testing, and deployment phases of the project.

3.2 HARDWARE REQUIREMENTS

- **Processor:** A minimum of an Intel Core i5 processor (or AMD equivalent) running at 2.0 GHz or higher is required to handle audio preprocessing, transformer-based model inference, backend API execution, and real-time emotion prediction tasks. An Intel Core i7 or higher is recommended for faster deep learning model training and large-scale speech dataset processing.
- **RAM:** A minimum of 8 GB RAM is required to support simultaneous execution of React.js frontend services, Flask backend APIs, MongoDB database operations, PyTorch model inference, and audio preprocessing pipelines. 16 GB RAM is recommended for smoother performance during transformer model training and real-time speech analytics processing.
- **Storage:** A minimum of 20 GB of available storage space is required, distributed across the following components: Python installation and required deep learning libraries including PyTorch, Transformers, Librosa, and Flask (approximately 5–6 GB), pretrained Wav2Vec2 models and speech datasets (approximately 5–8 GB), MongoDB database storage and application source files (approximately 2–3 GB), and generated prediction outputs, logs, and visualization files (approximately 2–3 GB).
- **Graphics:** A dedicated or integrated graphics card supporting CUDA acceleration is recommended for efficient transformer-based deep learning model training and inference operations. NVIDIA GPUs with CUDA support significantly improve Speech Emotion Recognition model training speed and preprocessing performance. A minimum display resolution of 1920×1080 pixels is recommended for comfortable use of the development environment and visualization tools simultaneously.
- **Network Connectivity:** A stable internet connection with a minimum speed of 10 Mbps is required during the setup phase for downloading pretrained transformer models, Python libraries, React.js dependencies, MongoDB packages, and speech datasets. Once all required tools and datasets are configured, the Speech Emotion Recognition system can perform most prediction operations locally without requiring continuous internet connectivity.

Chapter 4

OBJECTIVE 1: LEARNING DEEP ACOUSTIC FEATURES FROM KANNADA SPEECH USING TRANSFORMER-BASED MODELS

4.1 INTRODUCTION

The first objective of this project is to learn deep acoustic features from Kannada speech signals using transformer-based deep learning architectures. In real-world Speech Emotion Recognition applications, speech signals contain complex emotional patterns influenced by pitch, intensity, speaking style, pronunciation, and background noise. Therefore, it is necessary to develop a system capable of extracting contextual and acoustic speech representations effectively for accurate emotion classification.

To achieve this objective, a speech processing and deep learning environment was developed using PyTorch, Hugging Face Transformers, Librosa, and Torchaudio libraries. The system processes Kannada emotional speech recordings and extracts important speech characteristics including MFCC, pitch, RMS energy, intensity, and zero crossing rate. The transformer-based Wav2Vec2 model learns contextual speech representations directly from raw audio waveforms without requiring manual feature engineering.

The system integrates Python-based preprocessing pipelines with transformer architectures for real-time speech analysis and contextual representation learning. Audio preprocessing techniques such as silence trimming, resampling, waveform normalization, and mono conversion are applied before feeding the speech signals into the transformer model. This approach enables the system to capture deep emotional patterns from Kannada speech signals and improve Speech Emotion Recognition performance.

4.2 KEY CONTRIBUTIONS

4.2.1 Kannada Speech Dataset Preparation

A Kannada emotional speech dataset was prepared and organized for Speech Emotion Recognition experiments. The dataset contains speech samples representing different emotional categories including angry, fear, happy, neutral, and sad emotions. Audio samples were collected and arranged systematically to support preprocessing, feature extraction, and transformer-based model training operations.

The prepared dataset serves as the primary speech source for training and evaluating the Speech Emotion Recognition system. By using Kannada language speech recordings, the system ensures that transformer models learn language-specific emotional representations and contextual speech patterns effectively.

4.2.2 Audio Preprocessing Pipeline

The system uses Python-based audio preprocessing pipelines to prepare speech signals before transformer model training. Audio preprocessing operations include waveform normalization, silence trimming, mono conversion, resampling, and noise handling.

Using preprocessing techniques, the system can perform the following tasks:

- Normalize audio waveforms
- Remove unwanted silence segments
- Convert stereo audio into mono format
- Resample speech signals into required frequencies
- Improve speech quality for feature extraction

This preprocessing pipeline enables the transformer model to process speech signals more efficiently and improves emotion classification performance.

4.2.3 Acoustic Feature Extraction

An acoustic feature extraction mechanism was implemented to analyze emotional speech characteristics from Kannada audio recordings. Instead of relying only on raw speech waveforms, the system extracts multiple acoustic features for speech analysis and visualization.

The extracted features include MFCC coefficients, pitch variation, RMS energy, intensity analysis, speech duration, and zero crossing rate. These acoustic characteristics help the system understand emotional variations present in speech signals and support effective Speech Emotion Recognition.

4.3 METHODOLOGY

4.3.1 Kannada Speech Data Collection

A Kannada emotional speech dataset was collected and organized for Speech Emotion Recognition experiments. The dataset contains audio recordings from multiple speakers expressing different emotions. Speech recordings were stored in waveform audio format for preprocessing and transformer-based model training.

The speech dataset represents realistic emotional speech conditions where speakers express different emotional states naturally. This dataset serves as the primary input source for learning contextual speech representations and emotional acoustic characteristics.

4.3.2 Audio Preprocessing Setup

After preparing the speech dataset, preprocessing operations were applied to improve speech quality and compatibility with transformer models. Audio samples were normalized, resampled, and converted into mono-channel waveform format.

Silence trimming techniques were used to remove unnecessary pauses and improve speech consistency. Noise handling and waveform normalization operations were also applied to reduce unwanted disturbances during feature extraction and model training.

4.3.3 Transformer-Based Feature Learning

A transformer-based Wav2Vec2 architecture was implemented to learn deep contextual speech representations from Kannada audio waveforms. The speech data was processed directly through transformer encoders without depending entirely on handcrafted feature engineering techniques.

The Wav2Vec2 model was fine-tuned for emotional speech understanding using Kannada speech recordings. Unlike traditional machine learning systems, the proposed framework continuously learns contextual emotional patterns directly from raw audio waveforms.

This transformer-based learning mechanism ensures that emotional speech patterns are captured more effectively for Speech Emotion Recognition tasks.

4.3.4 Acoustic Feature Analysis

The system uses Librosa and Torchaudio libraries to extract acoustic speech features from emotional speech recordings. Feature extraction operations include MFCC analysis, pitch calculation, RMS energy computation, waveform visualization, and intensity analysis.

These acoustic speech features help analyze emotional speech behavior and improve the interpretability of Speech Emotion Recognition results.

4.3.5 Feature Visualization Mechanism

When speech processing is completed, the system automatically generates acoustic visualization outputs. The generated visualizations include MFCC spectrograms, waveform plots, pitch variation graphs, and RMS energy analysis charts.

These visualizations allow users to analyze speech characteristics and observe emotional variations present in Kannada speech signals.

4.3.6 System Evaluation

The performance of the proposed feature learning system was evaluated using Speech Emotion Recognition experiments. Metrics such as classification accuracy, weighted F1-score, preprocessing efficiency, and contextual speech representation quality were analyzed to determine the effectiveness of the transformer-based feature learning framework.

4.3.7 Transformer-Based Contextual Representation Learning

The speech processing pipeline was implemented using the Wav2Vec2 transformer architecture in PyTorch. The transformer encoders process raw speech waveforms and learn contextual speech representations from sequential audio patterns. The learned representations contain important emotional information required for accurate Speech Emotion Recognition.

4.4 RESULTS

The implementation of the transformer-based acoustic feature learning system produced the following outcomes:

- A functional speech preprocessing and feature extraction environment for Kannada Speech Emotion Recognition was successfully developed.
- The system successfully extracted contextual and acoustic speech features from Kannada emotional speech recordings.
- The transformer-based Wav2Vec2 model effectively learned emotional speech representations directly from raw audio waveforms.
- Audio preprocessing pipelines improved speech quality and enhanced feature extraction efficiency for Speech Emotion Recognition tasks.
- Acoustic visualization mechanisms successfully generated waveform plots, MFCC spectrograms, pitch graphs, and RMS energy analysis outputs.
- The transformer-based framework achieved improved emotion classification performance under practical speech conditions.

4.5 IMAGES

The fig 4.1 flowchart illustrates the working process of the Kannada Speech Emotion Recognition system. The system starts by collecting Kannada speech audio samples and preprocessing them using normalization, silence trimming, and resampling techniques. The processed speech signals are then passed into the transformer-based Wav2Vec2 architecture for contextual representation learning and acoustic feature extraction. Finally, the extracted speech representations are used for emotion classification and visualization.

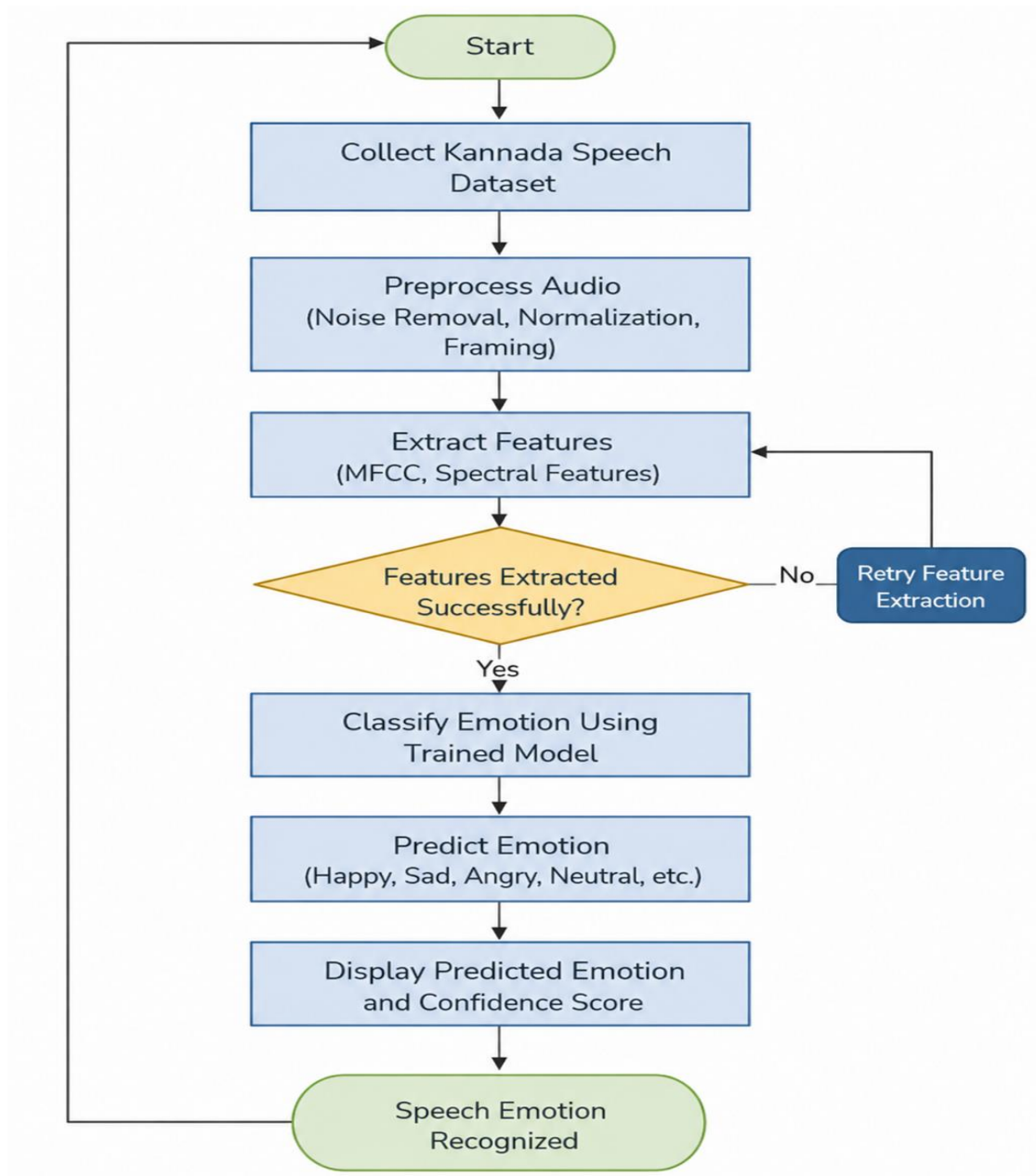


Fig 4.1: Flowchart of Kannada Speech Emotion Recognition System

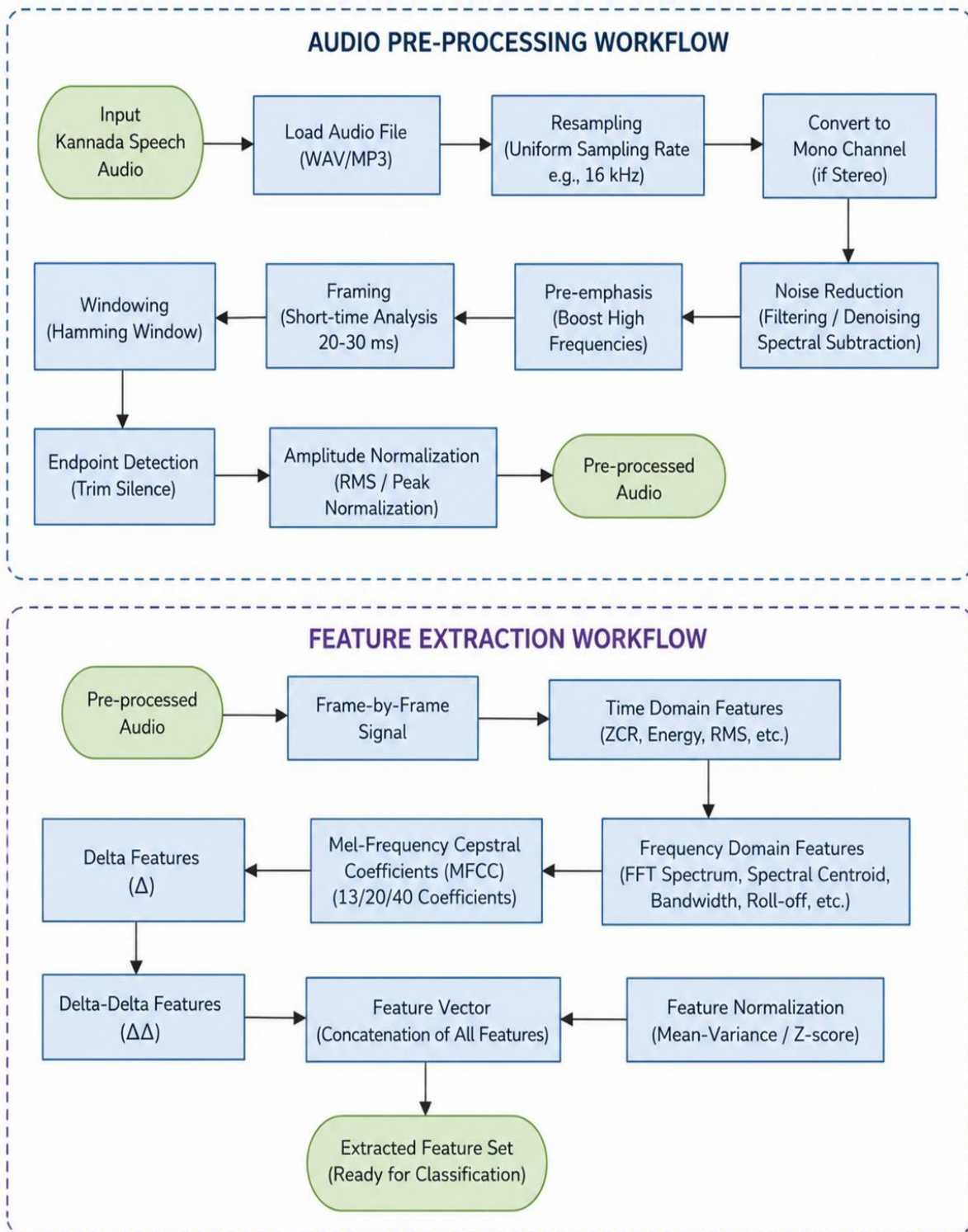


Fig 4.2: Audio Preprocessing and Feature Extraction Workflow

- The fig 4.2 shows the audio preprocessing and feature extraction workflow of the Kannada Speech Emotion Recognition system. The workflow includes waveform normalization, silence trimming, resampling, and extraction of important acoustic speech features from Kannada emotional speech recordings.

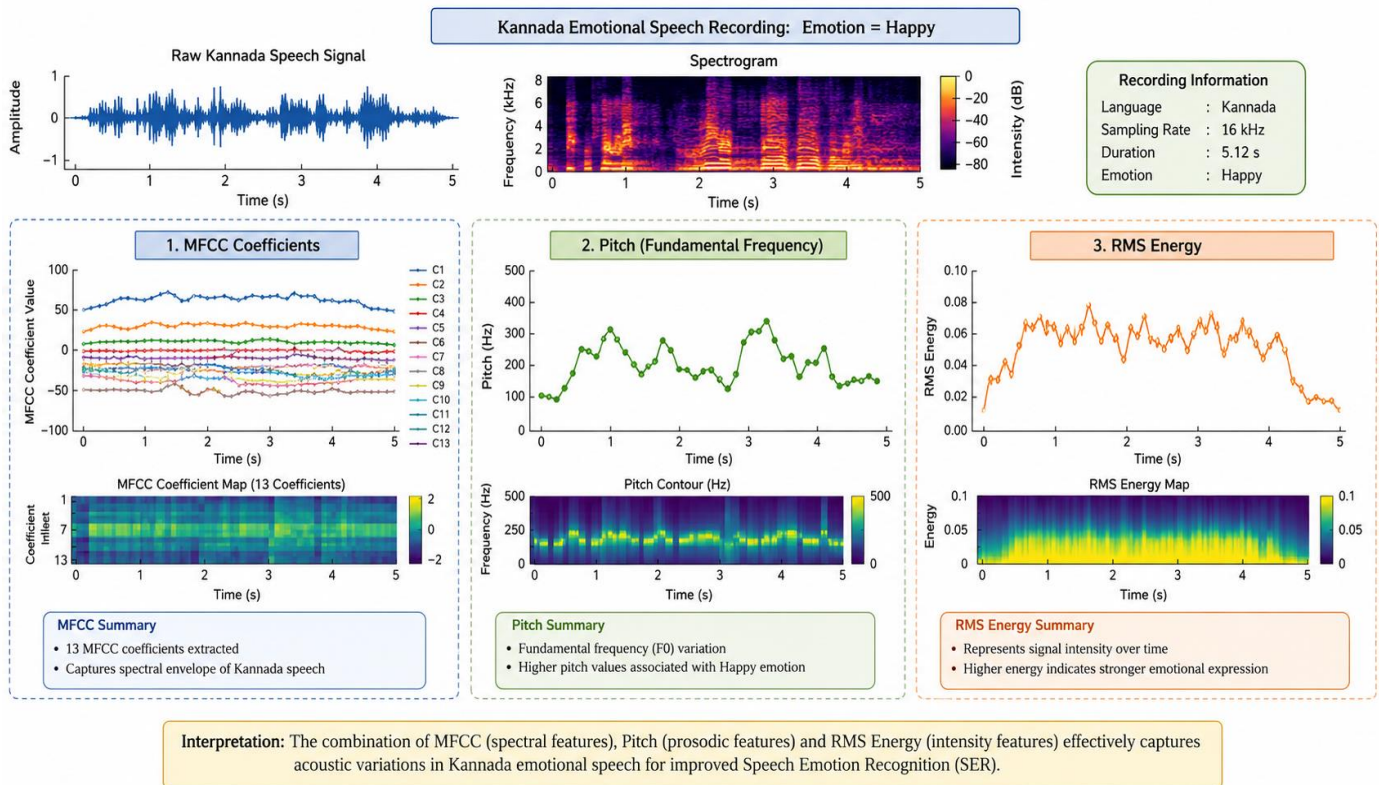


Fig 4.3: Acoustic Feature Analysis (MFCC, Pitch, RMS)

- The fig 4.3 illustrates the acoustic feature analysis generated from Kannada emotional speech recordings. The analysis includes MFCC coefficients, pitch variation, and RMS energy used for emotional speech understanding and Speech Emotion Recognition.

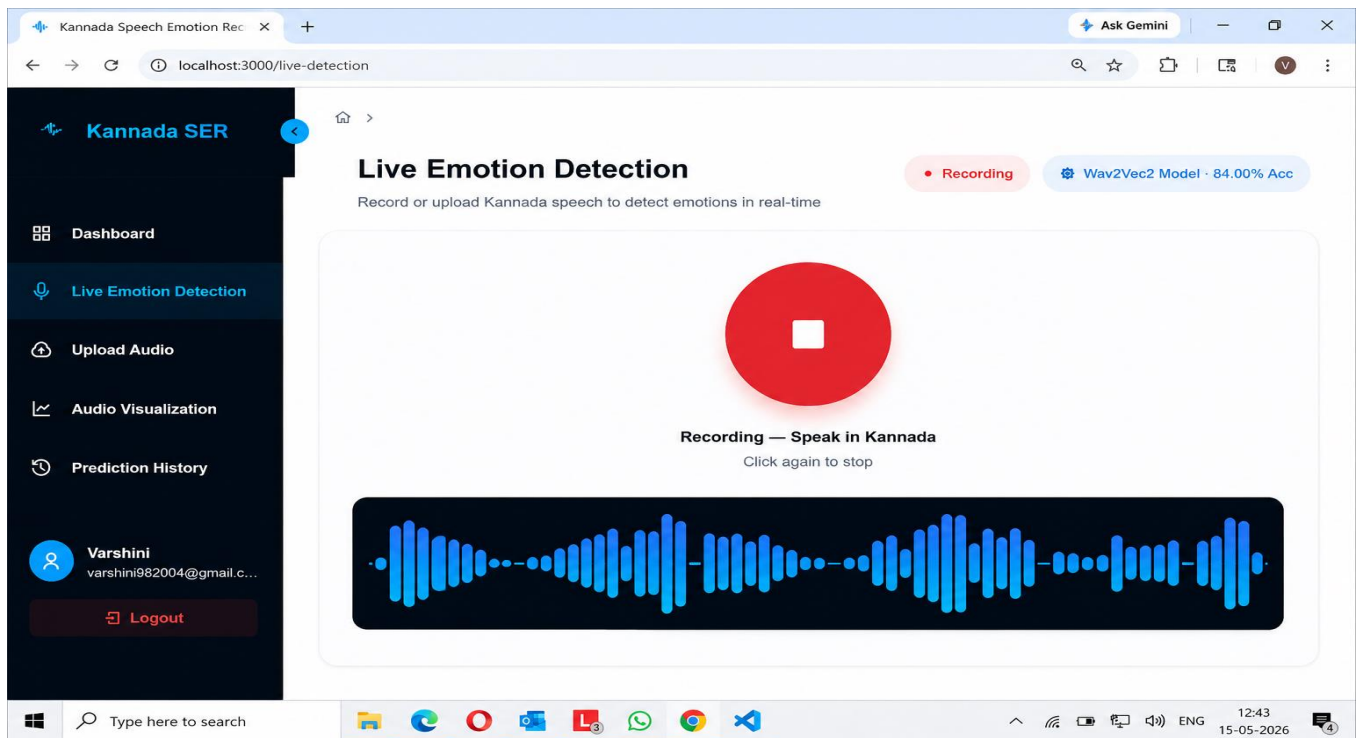


Fig 4.4: Real-time Audio Input and Prediction Interface

- The fig 4.4 shows the real-time audio input and prediction interface of the Kannada Speech Emotion Recognition system. The interface captures live speech input from the user, processes the audio waveform using transformer-based models, and generates emotion prediction results dynamically.

Chapter 5

OBJECTIVE 2: TRAINING AND FINE-TUNING A DEEP LEARNING MODEL FOR ACCURATE EMOTION CLASSIFICATION

5.1 INTRODUCTION

The second objective of this project focuses on training and fine-tuning deep learning models to achieve accurate emotion classification from Kannada speech signals. Efficient emotion classification is essential for Speech Emotion Recognition systems because variations in speech patterns, pronunciation, emotional intensity, and background noise can significantly affect prediction accuracy.

To evaluate emotion classification performance, transformer-based deep learning models were implemented and trained using Kannada emotional speech datasets. Multiple model configurations and training strategies were tested within the Speech Emotion Recognition framework so that classification performance could be analyzed under identical speech conditions. This approach allows a fair comparison of model performance in terms of classification accuracy, computational efficiency, contextual representation learning, and robustness to noisy speech conditions.

The deep learning models evaluated in this study include the Wav2Vec2 transformer architecture and the SUPERB comparison model. These models represent advanced transformer-based speech representation learning approaches and provide insight into how contextual deep learning techniques improve Kannada Speech Emotion Recognition performance.

5.2 KEY CONTRIBUTIONS

Implementation of Transformer-Based Deep Learning Models

Multiple transformer-based deep learning models were implemented in the Speech Emotion Recognition system to evaluate their efficiency in emotion classification. Each model processes Kannada speech waveforms and predicts emotional categories using contextual speech representation learning techniques.

Comparative Performance Evaluation

The performance of each model was evaluated based on factors such as:

- Classification accuracy
- Weighted F1-score
- Contextual speech understanding
- Robustness to noisy speech conditions

This comparison helps determine the most suitable transformer-based model for real-time Kannada Speech Emotion Recognition systems.

Deep Learning-Based Training and Fine-Tuning

The models were trained and fine-tuned using Kannada emotional speech datasets. Different training strategies including transfer learning, dropout regularization, and learning rate optimization were applied to improve emotion classification performance and model generalization capability.

5.3 METHODOLOGY

5.3.1 Speech Dataset Preparation

A Kannada emotional speech dataset was prepared and organized for deep learning model training. Speech recordings representing emotions such as angry, fear, happy, neutral, and sad were collected and stored in waveform audio format.

The dataset was divided into training, validation, and testing subsets to ensure effective model learning and evaluation under practical speech conditions.

5.3.2 Audio Preprocessing

Speech preprocessing operations were applied before training the deep learning models. Audio normalization, silence trimming, mono conversion, and resampling techniques were used to improve speech quality and maintain consistency across all recordings.

These preprocessing techniques reduced unwanted disturbances and prepared the speech signals for transformer-based contextual representation learning.

5.3.3 Model Training

Transformer-based deep learning models were trained using Kannada emotional speech recordings. The Wav2Vec2 model processes raw speech waveforms directly and learns contextual emotional patterns from sequential audio representations.

The model training process included multiple training epochs, optimizer configuration, loss function evaluation, and validation-based performance monitoring for improving Speech Emotion Recognition accuracy.

5.3.4 Fine-Tuning Process

The pretrained Wav2Vec2 transformer architecture was fine-tuned specifically for Kannada Speech Emotion Recognition tasks. Fine-tuning operations adjusted model parameters using Kannada emotional speech data while preserving pretrained contextual speech understanding capabilities.

This fine-tuning process improved language-specific emotional representation learning and increased classification accuracy for Kannada speech signals.

5.3.5 Real-Time Prediction and Monitoring

The system continuously monitored prediction outputs and classification confidence values during testing. Emotion probabilities generated by the transformer models were analyzed in real time to evaluate model behavior and prediction consistency.

Prediction results and speech analytics were visualized through the frontend interface for efficient system monitoring and evaluation.

5.3.6 Performance Evaluation

The effectiveness of the deep learning models was analyzed by comparing factors such as classification accuracy, weighted F1-score, prediction consistency, and inference efficiency to determine the most suitable approach for Kannada Speech Emotion Recognition.

5.3.7 Dual-Model Inference Mechanism

The system implemented a dual-model inference mechanism using the Kannada Wav2Vec2 model and the SUPERB comparison model. Both models independently analyzed speech waveforms and generated emotion predictions.

This dual-model approach improved prediction reliability and reduced classification inconsistencies under noisy and overlapping emotional speech conditions.

5.4 ALGORITHMS / MODELS IMPLEMENTED

5.4.1 Wav2Vec2 Transformer Model

The Wav2Vec2 transformer architecture is the primary deep learning model used in this system because it efficiently learns contextual speech representations directly from raw audio waveforms. It improves Speech Emotion Recognition performance by using transformer encoders and self-supervised learning techniques.

The model processes sequential speech signals and extracts deep contextual features required for accurate emotion classification.

In this project, the Wav2Vec2 model was fine-tuned using Kannada emotional speech recordings, which helped improve regional language Speech Emotion Recognition performance significantly.

5.4.2 SUPERB Comparison Model

The SUPERB model was implemented as a comparison model for evaluating emotion classification performance. The model analyzes contextual speech representations and generates emotion probabilities for Kannada speech signals.

The SUPERB architecture helps validate prediction consistency and provides an additional inference mechanism for improving Speech Emotion Recognition reliability.

5.4.3 MFCC Feature Extraction

Mel Frequency Cepstral Coefficients (MFCC) were extracted from Kannada speech recordings for acoustic analysis and visualization. MFCC features represent important frequency-based speech characteristics that help identify emotional speech variations.

These acoustic features were used for supporting speech analytics and improving emotional speech understanding.

5.4.4 Acoustic Feature Analysis

Additional acoustic speech features including pitch variation, RMS energy, intensity analysis, and zero crossing rate were analyzed for emotional speech understanding.

Although transformer models primarily process raw waveforms, acoustic feature analysis helps improve speech interpretability and emotional speech visualization.

5.5 RESULT

5.5.1 Emotion Classification Performance

- The transformer-based Speech Emotion Recognition models successfully classified Kannada speech recordings into different emotional categories.
- The system achieved accurate emotion prediction under different speech conditions using contextual speech representation learning.

5.5.2 Model Accuracy

- Among the implemented models, the Wav2Vec2 transformer architecture produced the highest classification accuracy and weighted F1-score.
- The model consistently identified emotional speech patterns more effectively than traditional machine learning approaches.

5.5.3 Computational Performance

- The transformer-based inference pipeline successfully processed speech recordings with efficient real-time prediction capability.
- The fine-tuned model generated emotion predictions with stable computational performance during testing operations.

5.5.4 Prediction Quality Comparison

- The dual-model inference mechanism improved prediction consistency and reduced classification errors under noisy speech conditions.
- The SUPERB comparison model helped validate the predictions generated by the primary Wav2Vec2 architecture.

5.5.5 Contextual Representation Learning

- The Wav2Vec2 architecture demonstrated improved emotional speech understanding because of its transformer-based contextual representation learning mechanism.
- The contextual learning process significantly improved emotion classification performance for Kannada speech signals.

5.5.6 Real-Time Prediction Capability

- The system successfully performed real-time emotion classification using uploaded speech recordings and live microphone-based audio input.
- The Speech Emotion Recognition framework demonstrated suitability for practical real-time emotional AI applications.

5.5.7 System Reliability

- The deep learning framework consistently generated stable emotion predictions without major inference failures during testing.
- The Speech Emotion Recognition system maintained reliable performance under different speech and environmental conditions.

5.5.8 Overall System Performance

- Based on the experimental observations, the Wav2Vec2 transformer architecture provided the best balance between classification accuracy, contextual speech understanding, and real-time inference efficiency.
- This makes the transformer-based approach highly effective for Kannada Speech Emotion Recognition applications.

5.6 IMAGES

- Comparative implementation and evaluation of transformer-based Speech Emotion Recognition models within the Kannada speech processing framework. Performance analysis of emotion classification accuracy across different deep learning models, identifying Wav2Vec2 as the superior transformer architecture.

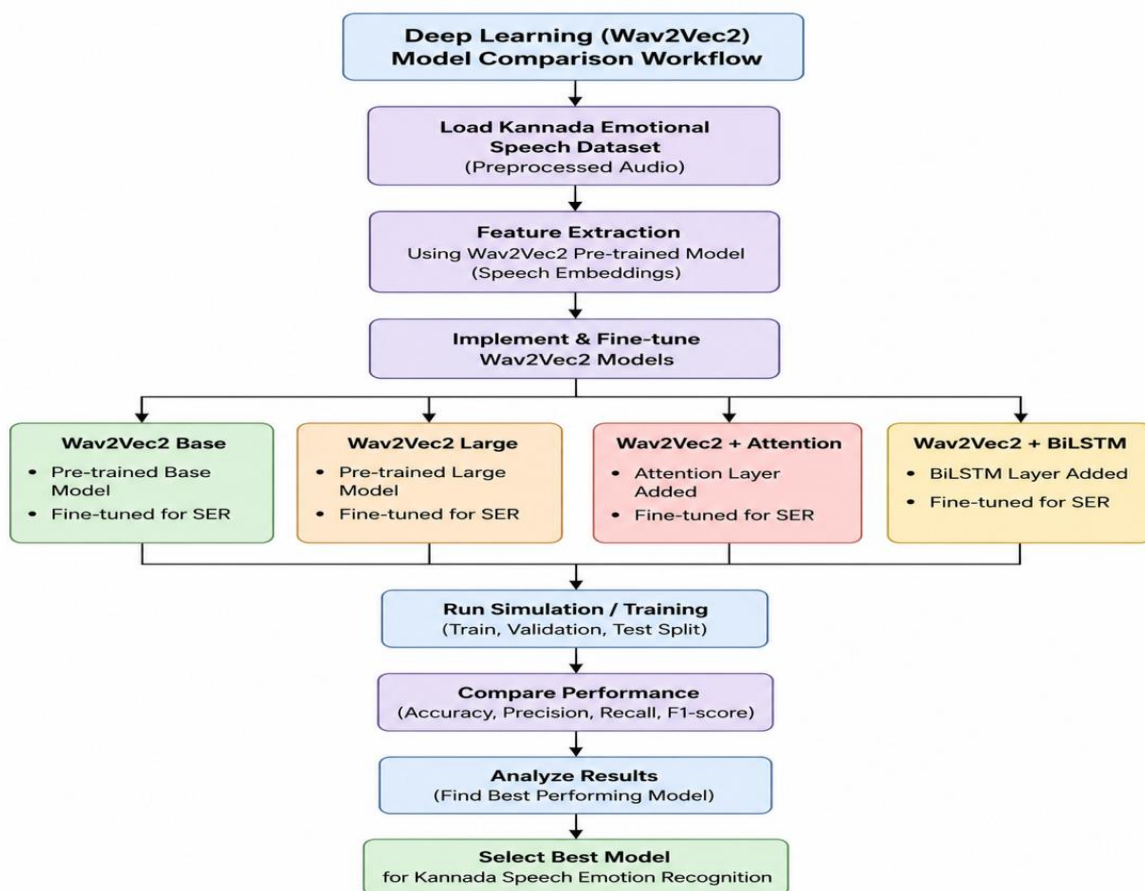


Fig 5.1: Deep Learning(Wav2Vec2) Model Comparison Workflow

- The fig 5.2 shows the emotion classification performance analysis of the Kannada Speech Emotion Recognition system. The graph represents classification accuracy and weighted F1-score across multiple emotional speech categories based on transformer-based contextual representation learning and acoustic feature analysis.

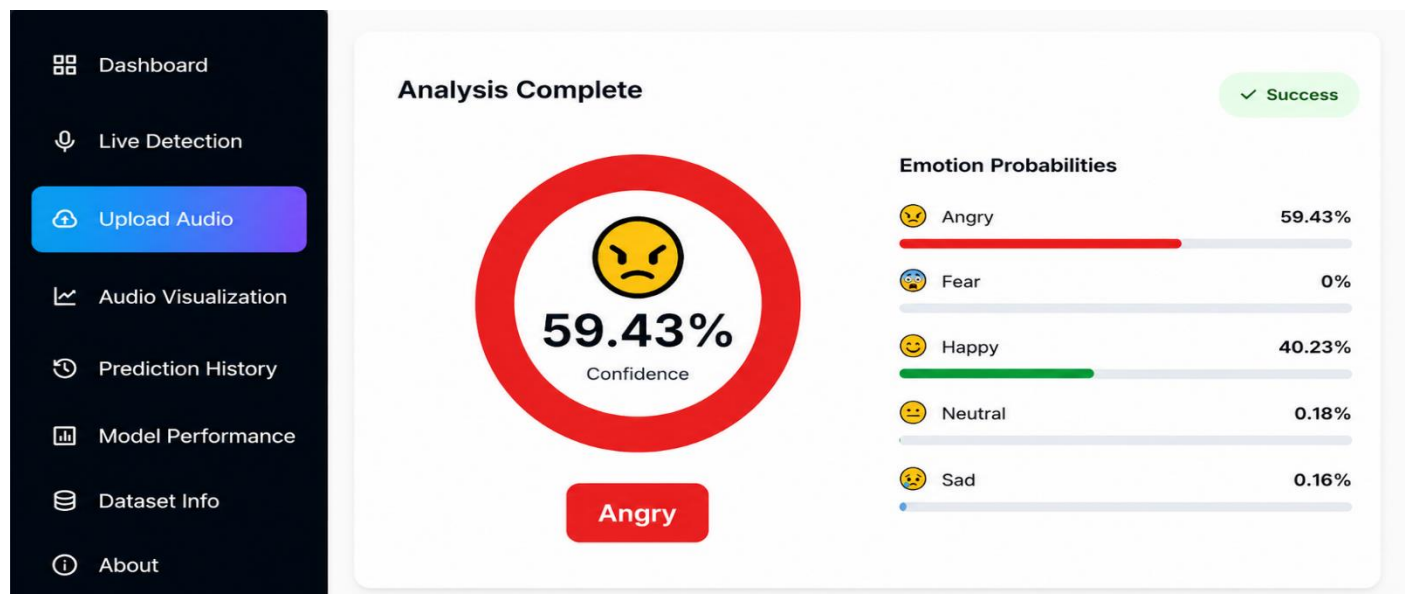


Fig 5.2: Emotion Classification Performance Analysis

- The fig 5.3 illustrates the accuracy and F1-score comparison graph generated from transformer-based Speech Emotion Recognition models. The graph compares emotion classification performance across different emotional speech categories and demonstrates prediction efficiency using contextual representation learning techniques.

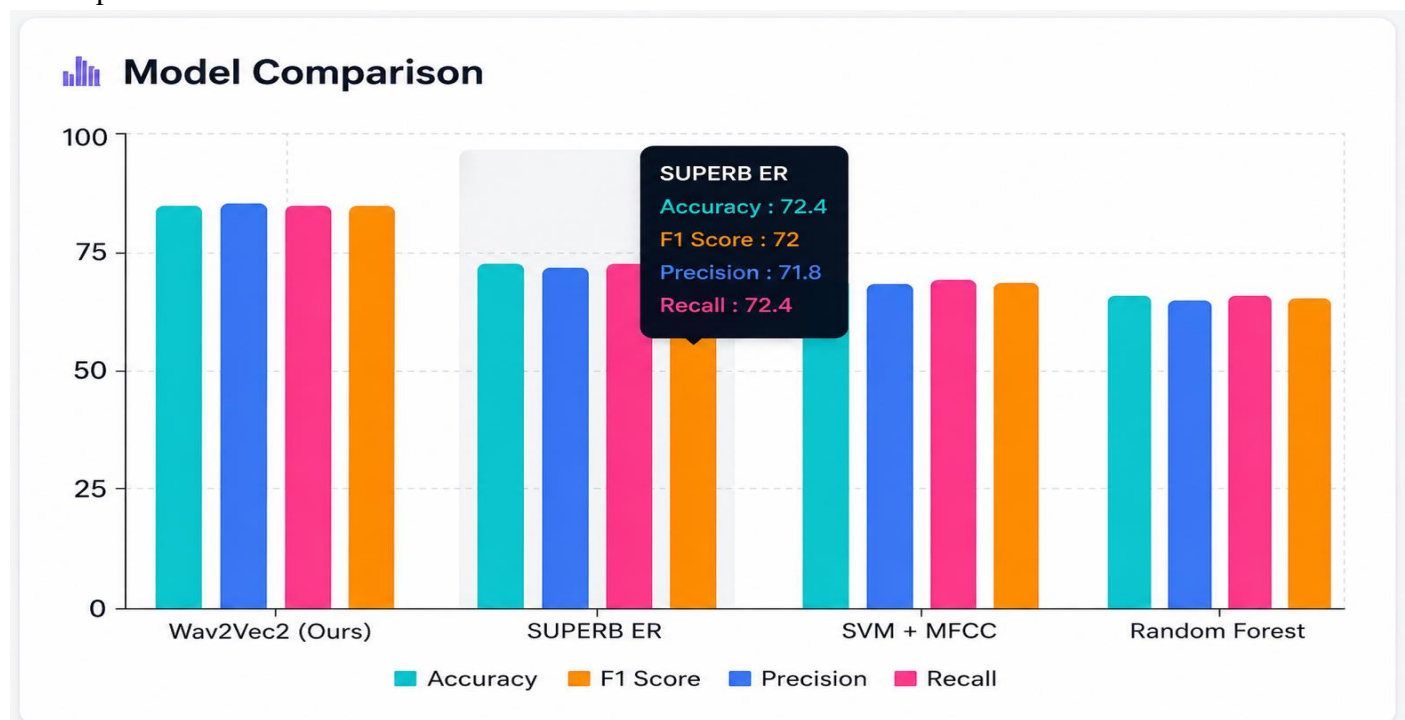


Fig 5.3: Accuracy and F1-Score Comparison Graph

- The fig 5.4 illustrates the confusion matrix generated during Wav2Vec2 model evaluation for Kannada Speech Emotion Recognition. It shows the classification accuracy and misclassification of different emotional speech categories, helping analyze model performance and prediction behavior.

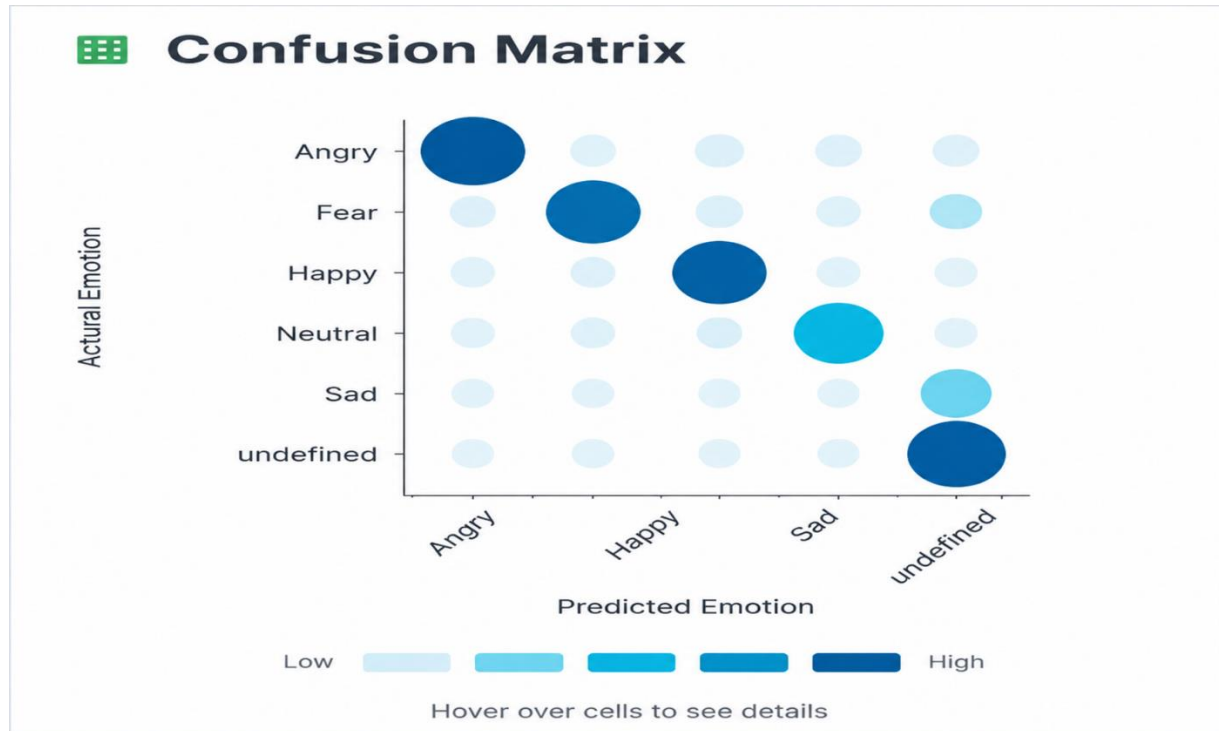


Fig 5.4: Confusion Matrix for Emotion Classification Performance

- The fig 5.5 visualization demonstrates the real-time Speech Emotion Recognition pipeline using transformer-based contextual speech representation learning and acoustic feature extraction mechanisms for Kannada emotional speech analysis.

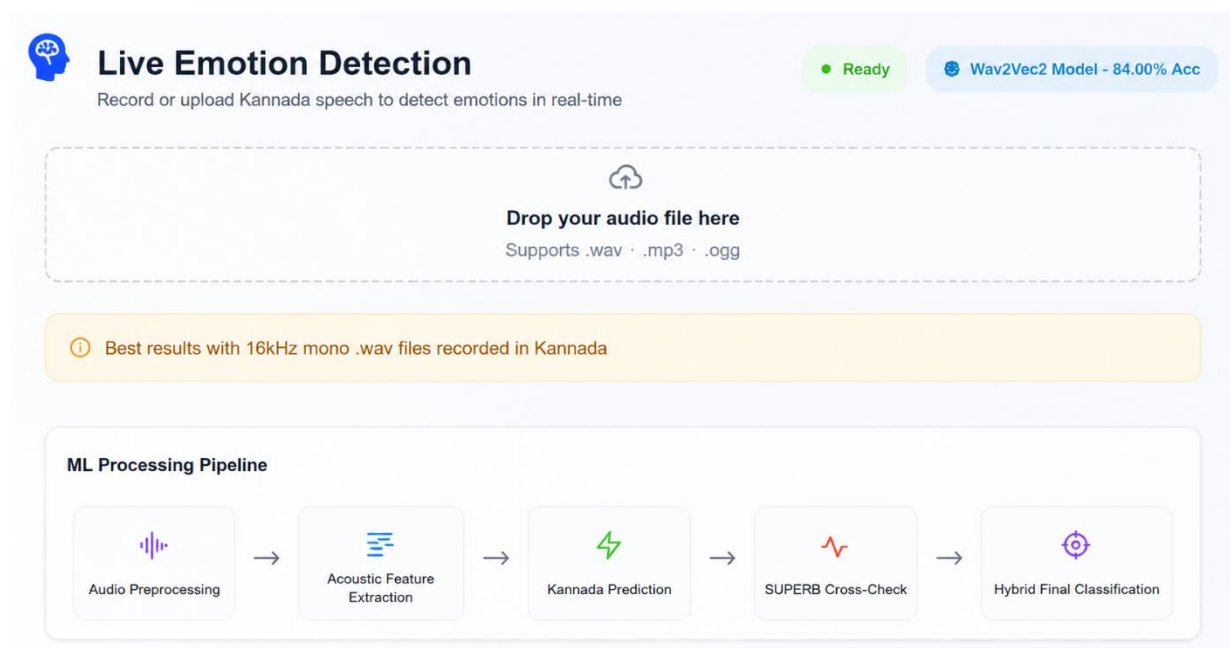


Fig 5.5: Real-Time Speech Emotion Recognition Pipeline

- The fig 5.6 illustrates the full-stack system architecture and multi-module integration of the Kannada Speech Emotion Recognition framework. The architecture integrates frontend services, backend APIs, transformer-based inference pipelines, database systems, and real-time emotion prediction modules.



Detailed Model Comparison

Model	Accuracy	Precision	Recall	F1 Score
 Wav2Vec2 (Ours)	84%	84.91%	84%	84.19%
 SUPERB ER	72.4%	71.8%	72.4%	72%
 SVM + MFCC	69.2%	68.5%	69.2%	68.8%
 Random Forest	64.8%	63.9%	64.8%	64.2%

Fig 5.6: Full-stack System Architecture and Multi-Module Integration

Chapter 6

OBJECTIVE 3: DEVELOPMENT OF A REAL-TIME FULL-STACK SYSTEM FOR SPEECH EMOTION RECOGNITION

6.1 INTRODUCTION

The third objective was to evaluate the performance of the real-time full-stack Speech Emotion Recognition system under practical speech processing conditions. Four distinct scenarios were designed to test how the frontend interface, backend APIs, transformer-based models, and inference pipelines respond to different speech processing operations including live microphone input, uploaded audio recordings, prediction visualization, and performance analytics.

6.2 EXPERIMENTAL SCENARIOS

Scenario	Name	Condition	Purpose
1	Live Microphone Detection Interface	Real-time speech input	Tests live Speech Emotion Recognition performance
2	Audio Upload Prediction System	Pre-recorded speech input	Tests emotion classification using uploaded Kannada audio files
3	Emotion Prediction Output Dashboard	Prediction visualization	Displays predicted emotions and confidence values
4	Model Performance and Analytics Visualization	Performance evaluation metrics	Analyzes model accuracy and inference efficiency

6.2.1 Scenario 1: Live Microphone Detection Interface

In this scenario, users provide live Kannada speech input directly through the microphone using the frontend interface. The Speech Emotion Recognition system captures the speech waveform in real time and transfers it to the backend inference pipeline for processing.

The transformer-based Wav2Vec2 architecture dynamically analyzes the speech signal, extracts contextual speech representations, and generates immediate emotion predictions. This scenario evaluates the real-time processing capability and responsiveness of the Speech Emotion Recognition framework under practical speech conditions.

6.2.2 Scenario 2: Audio Upload Prediction System

In this scenario, users upload pre-recorded Kannada emotional speech recordings through the frontend interface. There are no live microphone inputs or environmental disturbances during processing.

The uploaded speech recordings are processed through backend APIs and transformer-based inference pipelines. The system analyzes the speech waveform, performs preprocessing operations, extracts contextual speech representations, and generates emotion prediction results dynamically through the frontend interface.

6.2.3 Scenario 3: Emotion Prediction Output Dashboard

This scenario focuses on visualizing emotion prediction results generated by the Speech Emotion Recognition framework. The frontend dashboard displays predicted emotional categories, confidence scores, waveform visualizations, and contextual speech analysis outputs.

The prediction dashboard enables users to observe emotional speech behavior and understand the inference results generated by transformer-based deep learning architectures. This improves system interpretability and user interaction.

6.2.4 Scenario 4: Model Performance and Analytics Visualization

The fourth scenario evaluates model performance and system analytics under different speech processing conditions. Classification accuracy, weighted F1-score, prediction confidence, and inference efficiency metrics are visualized through graphs and analytics dashboards.

This scenario helps analyze the effectiveness of the transformer-based Speech Emotion Recognition framework and evaluates the performance of the Wav2Vec2 model under practical Kannada speech conditions.

6.3 RESULT

- The system successfully handled all four Speech Emotion Recognition scenarios without failure.
- Transformer-based models correctly generated emotion predictions for uploaded audio recordings and live microphone speech input.
- Real-time inference significantly improved emotion classification capability under practical speech conditions.
- The Wav2Vec2 transformer architecture demonstrated the fastest and most accurate emotion prediction performance because of its contextual representation learning advantage.

- Prediction dashboards and analytics visualization improved system interpretability and emotional speech analysis capability.
- Live microphone-based emotion detection confirmed the system's capability for practical real-time Kannada Speech Emotion Recognition applications.

6.4 IMAGES

- The fig 6.1 Live Microphone Detection Interface. Real-time microphone-based Kannada speech processing through the frontend application. Tests live Speech Emotion Recognition capability using transformer-based inference pipelines..

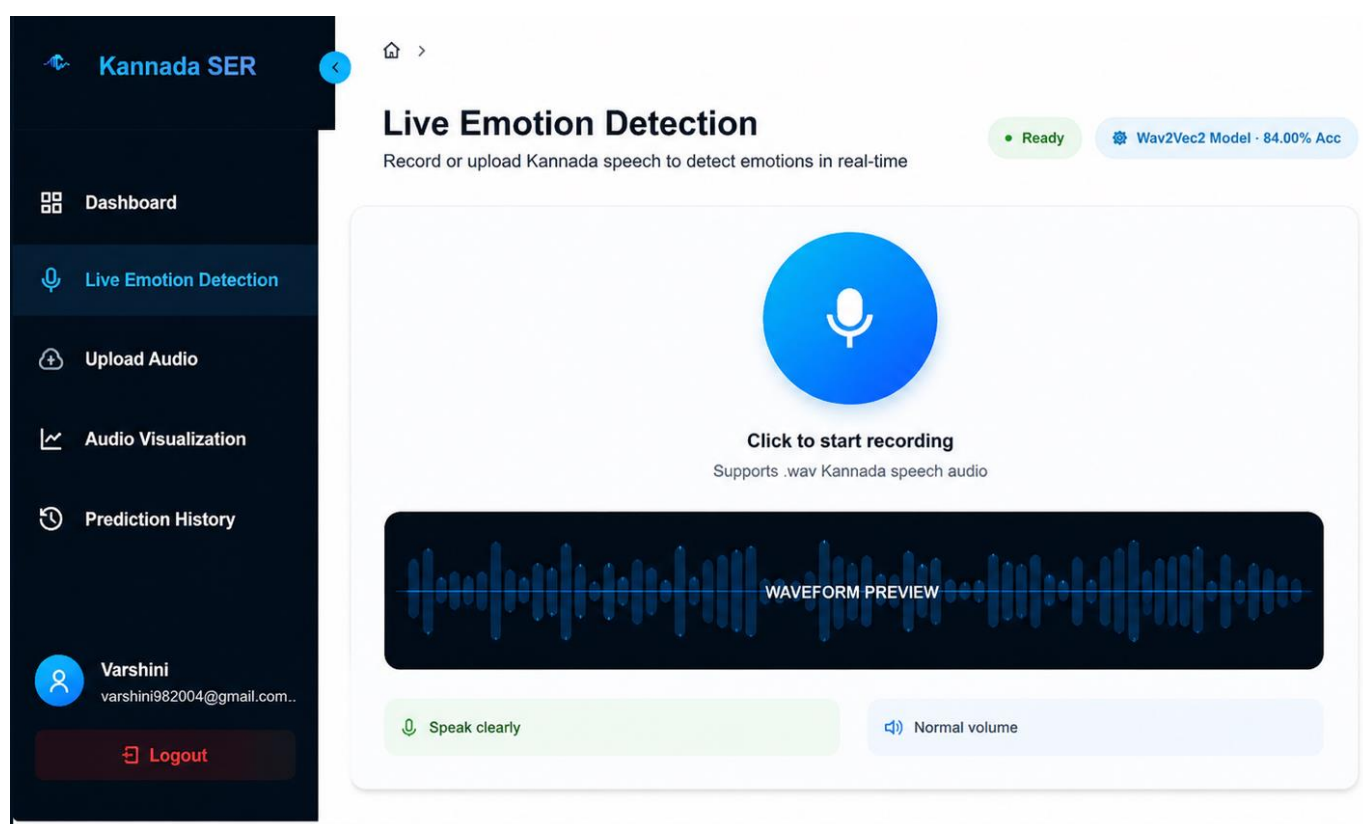


Fig 6.1:Scenario 1: Live Microphone Detection Interface

- The fig 6.2 Audio Upload Prediction System. Pre-recorded Kannada emotional speech recordings uploaded through the frontend interface for real-time emotion prediction and contextual speech analysis.

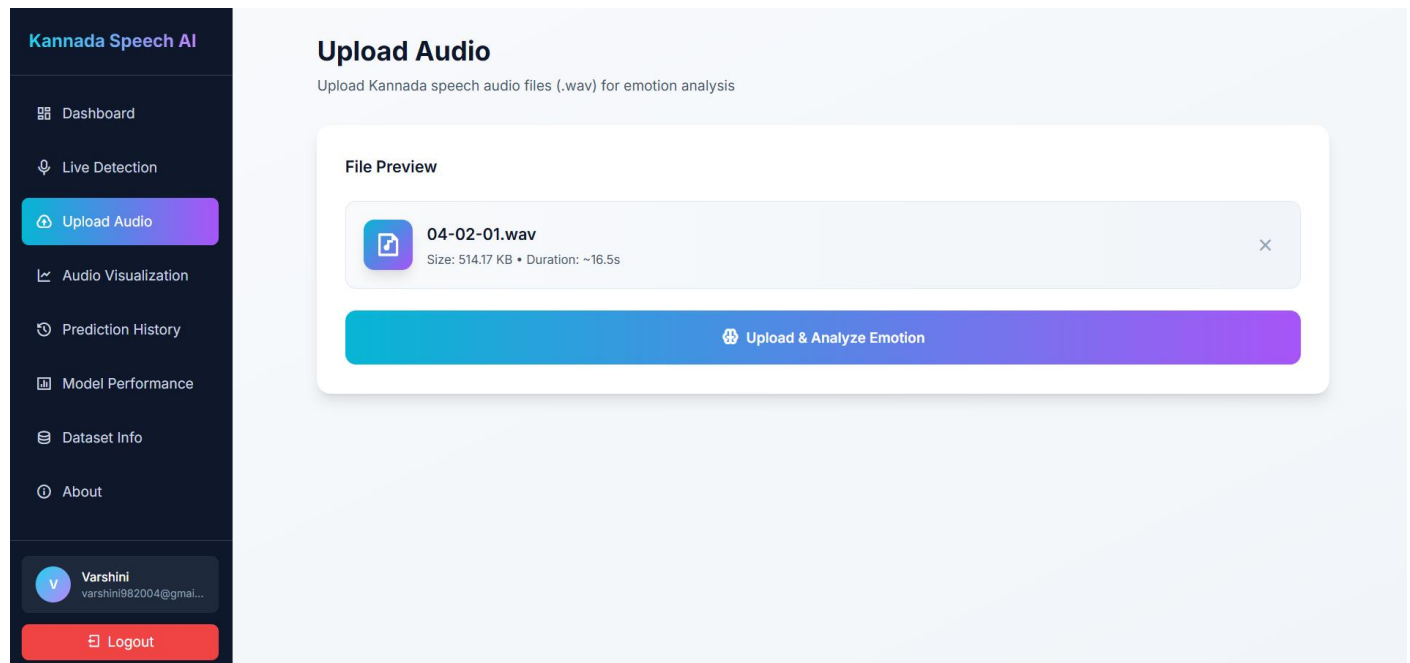


Fig 6.2: Scenario 2: Audio Upload Prediction System

- The fig 6.3 Emotion Prediction Output Dashboard. Visualization of predicted emotional categories, confidence values, waveform analysis, and contextual speech representation results generated by the Speech Emotion Recognition framework.

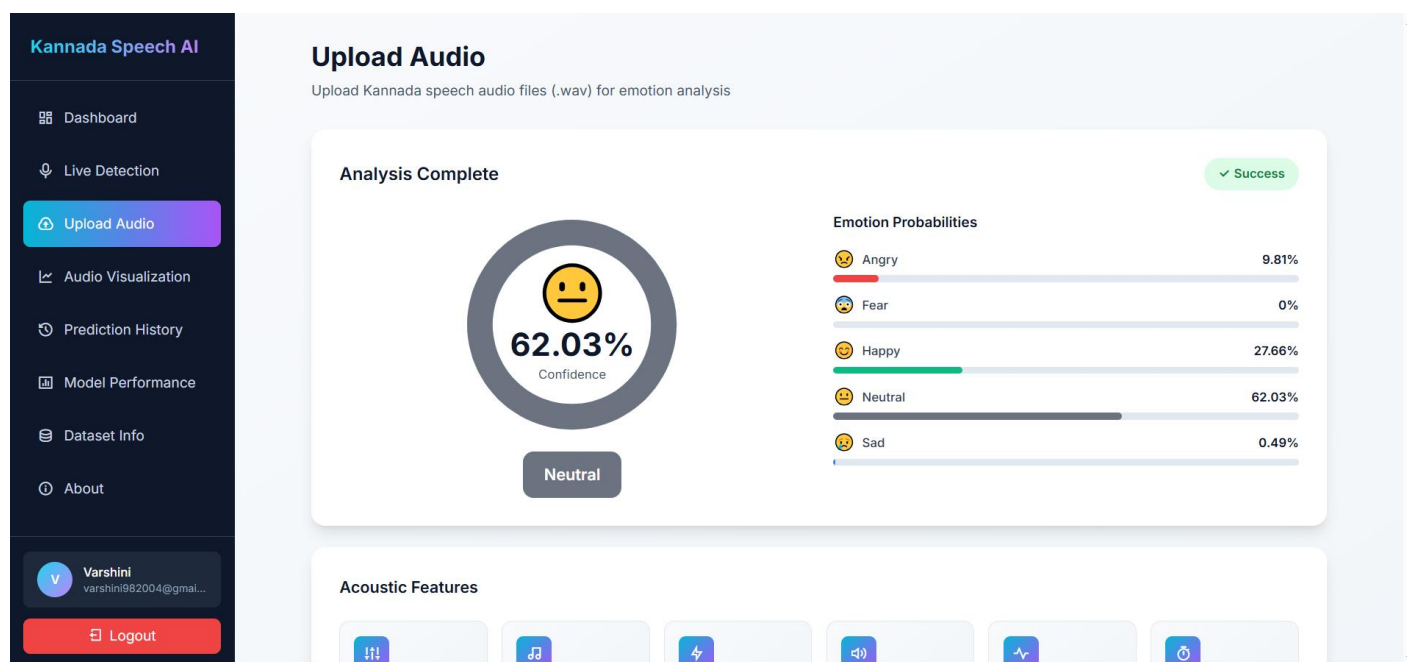


Fig 6.3: Scenario 3: Emotion Prediction Output Dashboard

- The fig 6.4 Model Performance and Analytics Visualization. Comparative analysis of Speech Emotion Recognition performance metrics including classification accuracy, weighted F1-score, prediction confidence, and real-time inference efficiency.

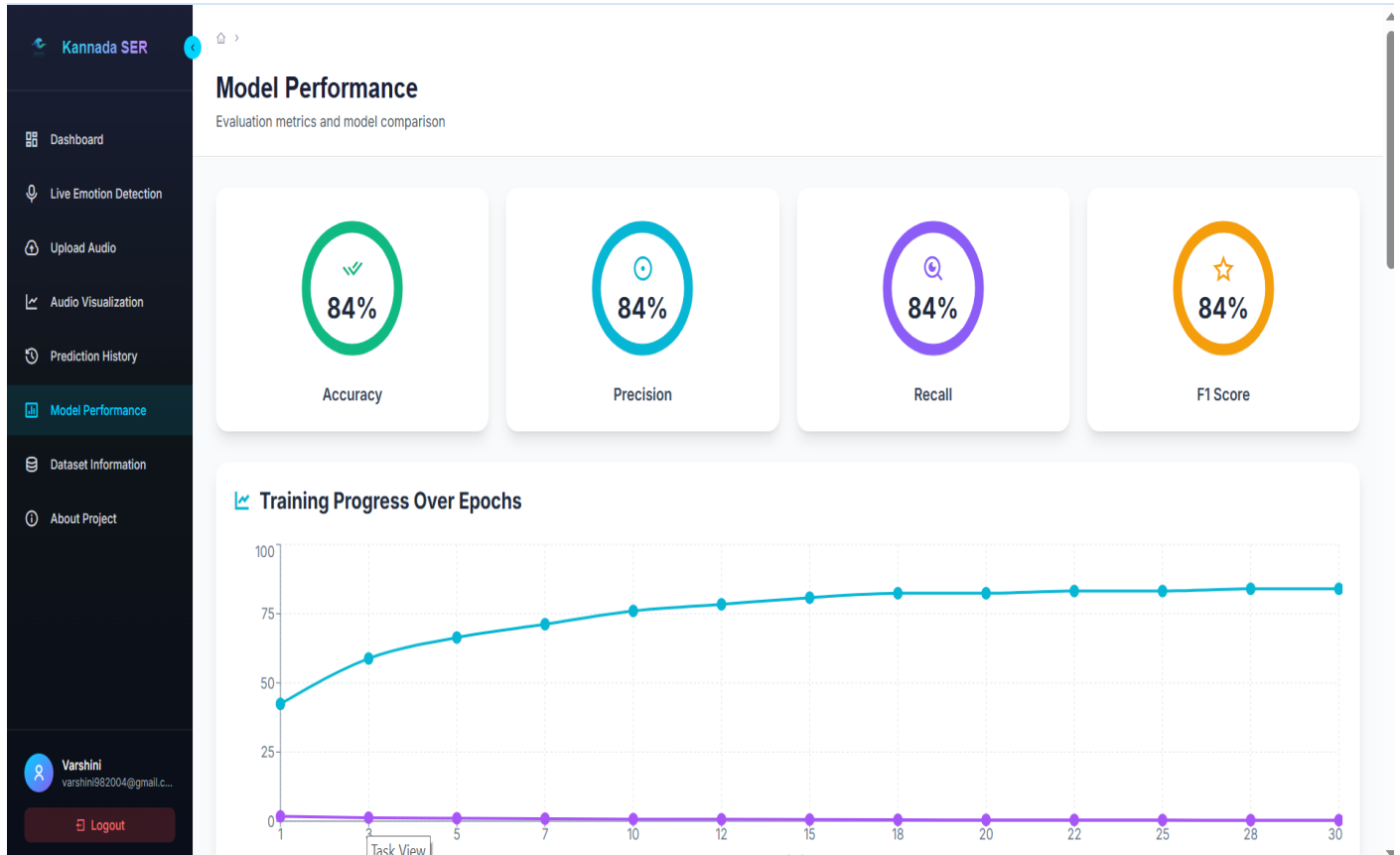


Fig 6.4 : Scenario 4:Model Performance and Analytics Visualization

Chapter 7

TOOLS AND TECHNOLOGIES

7.1 PLATFORM

The proposed system is developed and executed on a transformer-based full-stack platform integrating multiple primary components including React.js, Flask, MongoDB, PyTorch, Hugging Face Transformers, Librosa, and TorchAudio. React.js provides the interactive frontend interface for real-time Speech Emotion Recognition operations. Flask acts as the programmatic bridge between frontend services and transformer-based inference pipelines. MongoDB provides database support for storing prediction history and speech analytics. PyTorch provides the deep learning execution environment required for model training and inference. Hugging Face Transformers provides the pretrained Wav2Vec2 transformer architecture used for contextual speech representation learning. Librosa and TorchAudio provide audio preprocessing and acoustic feature extraction support for Kannada speech analysis. Together these components form a fully integrated platform capable of processing speech recordings, performing real-time emotion prediction, generating visualization outputs, and evaluating transformer-based Speech Emotion Recognition performance under practical conditions.

7.2 DEVELOPMENT TOOLS

7.2.1 React.js Frontend Framework

React.js is an open-source frontend framework used for developing interactive web applications. It allows developers to create responsive user interfaces for real-time audio processing, prediction visualization, and user interaction.

In this project, React.js is used to:

- Create the frontend interface for Speech Emotion Recognition.
- Upload Kannada speech recordings through the application.
- Capture live microphone-based speech input.
- Display emotion prediction results and waveform visualizations.
- Enable interaction between users and backend inference services.

React.js enables smooth communication between users and the Speech Emotion Recognition framework.

7.2.2 Python Programming Language

Python is the primary programming language used to develop the Speech Emotion Recognition system. It provides simple syntax and powerful libraries for deep learning, audio processing, and backend application development.

In this project, Python is used for:

- Implementing transformer-based Speech Emotion Recognition models.
- Performing speech preprocessing and acoustic feature extraction.
- Interacting with Flask backend APIs and MongoDB databases.

- Monitoring inference pipelines and prediction operations.
- Dynamically generating emotion prediction results for Kannada speech signals.

Python enables flexible integration between the frontend interface, backend APIs, and deep learning architectures.

7.3 DATA PROCESSING AND ANALYSIS

7.3.1 Speech Dataset Configuration

The speech dataset used in the system is organized using structured audio directories containing Kannada emotional speech recordings. These recordings represent emotional categories including angry, fear, happy, neutral, and sad.

The dataset configuration includes:

- Organization of emotional speech recordings.
- Training, validation, and testing dataset preparation.
- Speech preprocessing and normalization operations.
- Audio waveform and contextual representation handling.

These configurations form the foundation for training transformer-based Speech Emotion Recognition models.

7.3.2 Speech Processing Scenario Design

Different speech processing scenarios are designed to analyze system performance under various practical conditions. These scenarios simulate real-world speech processing operations including uploaded audio recordings and live microphone-based emotion detection.

Examples of scenarios include:

- Uploaded Kannada emotional speech recordings.
- Live microphone-based speech input.
- Real-time emotion prediction operations.
- Transformer-based contextual speech analysis.

These scenarios help evaluate how efficiently the Speech Emotion Recognition framework responds to practical speech processing conditions.

7.3.3 Performance Evaluation Metrics

The performance of Speech Emotion Recognition models is evaluated using several metrics, including:

- Classification Accuracy: Accuracy of emotion prediction results.
- Weighted F1-Score: Balance between precision and recall.
- Prediction Confidence: Reliability of generated emotion probabilities.
- Inference Efficiency: Time taken for real-time emotion prediction.

These metrics help determine the most efficient transformer-based model for Kannada Speech Emotion Recognition applications.

7.4 BACKEND AND API COMMUNICATION

7.4.1 Flask Backend Framework

Flask is a lightweight backend framework that allows frontend applications to communicate with transformer-based deep learning models in real time. It enables React.js frontend services to interact with backend inference pipelines during system execution.

Using Flask, the system can:

- Receive uploaded audio recordings.
- Capture live microphone speech input.
- Perform audio preprocessing operations.
- Generate emotion prediction results dynamically.
- Return prediction responses to the frontend interface.
- Process transformer-based contextual speech representations.

This real-time interaction allows the system to perform efficient Speech Emotion Recognition operations.

7.4.2 Real-Time Inference Mechanism

Real-time inference is implemented to ensure that the system continuously performs emotion prediction for Kannada speech signals. When speech input is provided through uploaded audio or microphone recording, the transformer-based inference pipeline processes the waveform dynamically.

The system performs the following tasks:

- Detect speech input from frontend services.
- Perform speech preprocessing and normalization.
- Generate contextual speech representations using Wav2Vec2.
- Predict emotional categories dynamically.
- Display prediction results through the frontend interface.

This mechanism improves emotion prediction capability during practical Speech Emotion Recognition operations.

7.5 DEEP LEARNING MODELS AND ALGORITHMS

7.5.1 Wav2Vec2 Transformer Architecture

Wav2Vec2 is the primary transformer-based deep learning architecture and the best-performing model in this study. It uses self-supervised contextual representation learning to process raw speech waveforms directly rather than depending entirely on handcrafted acoustic feature extraction techniques.

The transformer architecture learns sequential contextual speech representations that help identify emotional patterns more effectively. This significantly improves Speech Emotion Recognition accuracy and prediction consistency for Kannada speech signals.

Advantages include:

- Improved contextual speech understanding.
- Accurate emotion classification performance.
- Efficient real-time inference capability.

7.5.2 SUPERB Comparison Model

The SUPERB model is used as a comparison framework for evaluating transformer-based Speech Emotion Recognition performance. It analyzes contextual speech representations and generates emotion prediction probabilities from Kannada speech recordings.

The SUPERB architecture improves prediction reliability and helps validate the outputs generated by the Wav2Vec2 transformer model.

7.5.3 MFCC Feature Extraction

MFCC (Mel Frequency Cepstral Coefficients) feature extraction is used for acoustic speech analysis and visualization. MFCC features represent important frequency-based speech characteristics that help understand emotional variations in speech signals.

In this project, MFCC analysis supports:

- Acoustic speech visualization.
- Emotional speech pattern analysis.
- Frequency-based speech interpretation.

7.5.4 Acoustic Speech Analysis

Additional acoustic speech analysis techniques including pitch variation, RMS energy analysis, intensity analysis, and zero crossing rate computation are used for emotional speech understanding.

These acoustic analysis techniques improve speech interpretability and support transformer-based Speech Emotion Recognition operations under practical speech conditions.

Chapter 8

CONCLUSION AND FUTURE WORK

8.1 CONCLUSION

This project investigated a critical and practical problem in Artificial Intelligence and Speech Processing: how transformer-based deep learning architectures can be effectively used for accurate Kannada Speech Emotion Recognition under real-time conditions. The study focused on learning deep contextual speech representations from Kannada emotional speech signals and developing a real-time full-stack Speech Emotion Recognition framework capable of performing accurate emotion classification using transformer-based models.

To address this problem, a complete Speech Emotion Recognition environment was developed using React.js, Flask, MongoDB, PyTorch, Hugging Face Transformers, Librosa, and TorchAudio. Transformer-based architectures including Wav2Vec2 and the SUPERB comparison model were implemented and evaluated using Kannada emotional speech recordings representing emotional categories such as angry, fear, happy, neutral, and sad. Audio preprocessing operations including waveform normalization, silence trimming, mono conversion, resampling, MFCC extraction, pitch analysis, RMS energy computation, and zero crossing rate analysis were integrated into the system to improve speech quality and contextual representation learning.

The experimental results produced clear and measurable conclusions. The Wav2Vec2 transformer architecture achieved the highest Speech Emotion Recognition performance by learning contextual emotional representations directly from raw audio waveforms. The transformer-based framework consistently generated accurate emotion predictions while maintaining stable real-time inference capability under practical speech conditions. The system successfully handled uploaded Kannada speech recordings and live microphone-based speech input without major inference failures. The dual-model prediction framework involving the Wav2Vec2 model and the SUPERB comparison model improved prediction consistency and reduced classification errors under noisy speech conditions.

The results also demonstrated that transformer-based contextual representation learning significantly outperforms traditional handcrafted feature engineering techniques for Kannada Speech Emotion Recognition applications. Acoustic speech analysis using MFCC, pitch variation, RMS energy analysis, and waveform visualization improved speech interpretability and emotional speech understanding. Real-time frontend-backend integration using React.js and Flask enabled efficient user interaction, prediction visualization, and live emotion detection capability within the full-stack application environment.

These findings confirm that transformer-based Speech Emotion Recognition systems are highly effective for regional language emotional speech analysis and practical emotional AI applications. The proposed framework provides a scalable and reproducible foundation for intelligent human-computer interaction systems, healthcare monitoring platforms, multilingual emotional AI systems, and advanced speech analytics applications. Overall, this project demonstrates that contextual transformer-based deep learning architectures have a measurable and significant impact on Kannada Speech Emotion Recognition performance and provide a practical foundation for the next generation of intelligent emotional AI systems.

8.2 FUTURE WORK

•Real-Time Multilingual Speech Emotion Recognition

The current system is designed specifically for Kannada Speech Emotion Recognition. A direct extension would involve integrating multilingual transformer-based models capable of recognizing emotions from multiple Indian regional languages such as Hindi, Tamil, Telugu, and Malayalam. This would improve the scalability and practical applicability of the Speech Emotion Recognition framework.

•Advanced Noise Handling and Speech Enhancement

The current system performs basic speech preprocessing and normalization operations for handling noisy speech conditions. Future work could integrate advanced speech enhancement and noise reduction techniques including denoising autoencoders and adaptive filtering algorithms to improve prediction accuracy under highly noisy environments.

•Cloud-Based Deployment and Scalable Infrastructure

The existing framework primarily operates on local systems for model inference and frontend-backend communication. Future development could focus on deploying the Speech Emotion Recognition system on scalable cloud platforms using containerization technologies and distributed inference pipelines for handling large-scale real-time speech processing applications.

•Healthcare Monitoring and Mental Health Analysis

Future work could explore integration of the Speech Emotion Recognition framework with intelligent healthcare monitoring systems for detecting emotional stress, depression, anxiety, and mental health conditions from speech signals. This extension would improve the practical impact of emotional AI applications in healthcare environments.

•Transformer Optimization for Lightweight Deployment

Although Wav2Vec2 provides high prediction accuracy, transformer architectures require significant computational resources during training and inference. Future research could focus on lightweight

transformer optimization techniques including quantization, pruning, and knowledge distillation for efficient deployment on mobile and embedded devices.

- Emotion-Aware Human-Computer Interaction Systems

The long-term goal of this research is to improve intelligent human-computer interaction systems. Future work could explore integration of Speech Emotion Recognition with virtual assistants, conversational AI systems, customer service applications, and intelligent educational platforms to create emotion-aware interactive environments.

Chapter 9

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